Sensation and Perception

INVITED ADDRESS

IA107
Human brain network for visual motion processing

Hiroshi Ashida *Kyoto University, Japan*

Motion processing is an essential function of vision for many animals including humans, both for their own navigation and for detection of other moving targets. It has therefore been extensively studied across research areas of psychology, physiology, computer science, etc, but there remain many questions that are yet to be investigated. One such question is how the local motion signals are integrated and used for navigation together with other senses. Here I will introduce some results from our laboratory on the network of human brain areas underlying detection and integration of visual motion information. The following topics will be included :1) fMRI on the anomalous motion illusion (Ashida, et al., 2012, NeuroImage), 2) fMRI on the processing of complex optic flow and perceived vection (Uesaki and Ashida, 2015, Frontiers in Psychology, and 3) Visual-vestibular connection revealed by diffusion MRI (Uesaki, et al., in prep).

IA108
The treachery of images: Why the brain responds differently to real object than photos

Jody C Culham (1), Jacqueline C Snow (2), Theresa M Gerhard (3), Gudrun Schwarzer (3) *1. Western University, Canada; 2. University of Nevada Reno, United States of America; 3. Justis Liebig University Giessen, Germany*

Psychologists and neuroimagers commonly study perceptual and cognitive processes using images because of the convenience and ease of experimental control they provide. However, real objects differ from pictures in many ways, including the availability and consistency of depth cues and the potential for interaction. Across a series of neuroimaging experiments, we have shown that the brain responds differently to real objects than pictures, both in terms of the level of activation and the pattern. Moreover, in a developmental study, we have found that infants prefer to look at real objects compared to pictures, even after they have habituated to the real object. Taken together, these results suggest that real objects are more engaging, both perceptually and neurally, and open up new research directions to better understand which aspects of real objects drive these effects.

IA109
Understanding the perceptual experience of pain: an experimental approach

Patrick Haggard *University College London, United Kingdom*

Pain is a cardinal human experience, with undeniable impact on quality of life. Recent research advances are clarifying the molecular and neural mechanisms of pain, yet basic psychological questions about pain experience remain underinvestigated. In this talk, I will consider recent developments in the experimental psychology of pain. First, new techniques for delivering noxious stimulation have improved understanding of different sensory pathways linked to pain. Second, appropriate psychophysical techniques can improve pain measurement. These improvements enable more rigorous investigation of the brain basis of pain, and the different factors that influence pain perception. While clinical interest has traditionally focussed on chronic pain, studies of acute experimental pain provide a promising tool for scientific understanding of pain as a general class of mental state. I will therefore describe how a detailed understanding of pain perception may help in developing therapies that reduce or manage pain.

IA110
Motion illusions in stationary images

Akiyoshi Kitaoka *Ritsumeikan University, Japan*

Motion illusions in stationary images are classified into three major categories depending on whether the illusory motion is rendered (1) by retinal motion of the image, (2) along particular repetitive patterns, or (3) by other factors. The first category consists of two subcategories, one showing illusory motion in the same direction as retinal motion (e.g., fluttering heart illusion) and the other displaying it in different directions (e.g., drifting spines illusion). The second category includes the extended family of the Fraser-Wilcox illusion that is characterized by strong illusion in the peripheral vision and no illusion in the central vision (e.g., Rotating snakes illusion). There are other types that show illusion in the central vision as well (e.g., drifting arrows illusion). The third category includes a variety of op effects (e.g., scintillating grid illusion). The roles of color, eye movements or adaptation levels in these illusions are discussed.

IA111
Processing time with our auditory system

Christian Lorenzi *Ecole normale superieure, France*

Debate on how speech information is represented in the auditory system has revolved around the role of two neural/perceptual features encoding the temporal modulations of the acoustic signal (the "temporal envelope", ENV, and "temporal fine structure", TFS), their relative contribution to intelligibility and how that might be degraded by lesions to the peripheral and central auditory system. We will review psychophysical studies that investigated the development of ENV/TFS perception, the effects of cochlear and central lesions, and the relationship between ENV/TFS perception and speech intelligibility. Our results suggest that: i) the processing of ENV and TFS is "functional" by 6 months, and fine-tuned by language exposure between 6 and 10 months, ii) ENV is more important for speech identification, whereas TFS is more important for the segregation of competing sound sources, iii) reduced ability to process ENV and/or TFS explains deficits typically associated with cochlear and central damage and ageing.

INVITED SYMPOSIUM

IS091
Sensory substitution

Organizer: Malika Auvray *ISIR, CNRS, France*

Session Abstract:

Sensory substitution devices convert stimuli that are normally accessed through one sensory modality (e.g., vision) into stimuli accessible through another sensory modality (e.g., touch or audition). Studies conducted with these devices revealed an important structural and functional plasticity of the central nervous system. From a structural point of view, it has been shown that practice with visual-to-tactile and visual-to-auditory devices results in increased activation in the blind's visual cortex. From a functional point of view, users can build a perceptual space that possesses visual characteristics. Lectures of this symposium will rely on speakers from different recognized international research groups providing their novel findings involving brain imaging data, behavioral results, the development of novel coding and interfaces, and the theoretical frameworks underlying this field of research.

The Rewired Blind Brain

Maurice Ptito, Ron Kupers *Universities of Montreal and Copenhagen, Denmark*

Vision plays a central role in how we represent and interact with the world around us. Roughly 50% of the cerebral cortex in monkeys and 20-30% in humans are devoted to vision. The loss of vision, either at birth or later in life, must therefore, have profound consequences on brain organization. In this presentation, we will illustrate the anatomical reorganization that takes place in the blind brain using various neuroimaging and electrophysiological

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methods. We will show that the visual system of the blind, although atrophied, largely contributes to the funnelling of non-visual information to the occipital cortex. We will highlight here that tactile information is relayed to the visual cortex not only by cortico-cortical projections but also by an ascending thalamocortical pathway that reaches the visual cortex via the optic radiations. We conclude that absence of vision does not lead to a disabled brain but to a differently abled brain.

Investigating the spatial perspectives that are adopted by sighted and blind people to interpret patterns of tactile stimulation

Gabriel Arnold, Malika Auvray *Université Pierre et Marie Curie, France*

Different spatial perspectives can be taken on tactile stimuli. The spatial coordinates (left, right, top, and bottom) of tactile stimuli provided by a tactile interface and, consequently, the interpretation of these stimuli, depend on the perspective that is taken. We used a tactile recognition task of ambiguous symbols (the letters b, d, p, and q) presented on the trunk, in order to investigate which perspectives sighted and blind people adopt spontaneously. The results revealed that observers have a natural perspective, which can be trunk-centred, head-centred, or decentred. Sighted people predominantly adopted a trunk-centred perspective, as if the symbols were projected forward in front of them, whereas blind people predominantly adopted a head-centred perspective, as if the symbols were perceived on their body surface. The perspective that is taken constitutes a crucial cue to the ability of attributing tactile stimulation to external objects when using a visuo-tactile substitution device.

Substituting Time for Space

Vincent Hayward *Université Pierre et Marie Curie, France*

In sensory substitution, the codes that are used to map features accessible in one modality into accessible properties of a second modality are often not intuitive. A pre-existing mapping which has been observed in many species is that of associating spatial separation to phase delay. We created a device that produces delayed tactile pulses monotically separated in time according to data acquired by a hand-held (or wearable) optical range finder. Positive results have been obtained in the task of approaching unseen walls without explicit instructions about the nature of the underlying mapping, that is, the mapping was used spontaneously. We will present results of the application of this training-free device when roaming amidst sparse unseen obstacles.

Origins of task-specific sensory-independent organization in the visual and auditory brain: neuroscience evidence, open questions and clinical implications

Amir Amedi (1,2), Benedetta Heimler (1), Ella Striem-Amit (3) *1. The Hebrew University of Jerusalem, Israel; 2. Sorbonne Universite's, Universite' Pierre et Marie Curie (UPMC), Universite' Paris 06, Institut de la Vision, Unite' Mixte de Recherche en Sante' (UMRS) 968, France; 3. Harvard University, United States of America*

Evidence of task-specific sensory-independent (TSSI) plasticity from blind and deaf populations has greatly contributed to our understanding of brain organization. However, the principles determining the origins of this plasticity remain unclear. I will present recent data suggesting that a combination of the connectivity bias and sensitivity to task-distinctive features might account for TSSI plasticity in the sensory cortices as a whole, from the higher-order occipital/temporal cortices to the primary sensory cortices. I will discuss current theories and evidence, open questions and related predictions. Finally, I will present preliminary results suggesting the way predictions coming from TSSI plasticity can be eventually implemented within rehabilitation programs to maximize the efficiency of visual and auditory recovery.

IS092
Time in the brain

Organizers: Simon Grondin (1), Shozo Tobimatsu (2), Emi Hasuo (3) *1. Université Laval, Canada; 2. Kyushu University, Japan; 3. Aoyama Gakuin University/Japan Society for the Promotion of Science, Japan*

Discussant: Yoshitaka Nakajima Department of Human Science, Kyushu University, Japan

Session Abstract:

Perceiving the passage of time is fundamental for our everyday activities, such as playing sports, enjoying music, and having a conversation. Time cannot be perceived directly, and must be reconstructed in the brain from pieces of information obtained through various sensory organs. The present symposium will consist of a short introduction by Simon Grondin (Laval University, Canada) and four talks given by Emi Hasuo (Aoyama Gakuin University), Tsuyoshi Kuroda (Shizuoka University), Takako Mitsudo (Kyushu University), and Giovanna Mioni (University of Padova, Italy), followed by a discussion session by Yoshitaka Nakajima and Shozo Tobimatsu (Kyushu University). We will cover topics in time perception from basic psychophysical studies (Hasuo and Kuroda) to electrophysiological studies (Mitsudo) and clinical studies with patients (Mioni), and show how using simple stimulus patterns and observing the behavioral and/or neural responses to them carefully can reveal how the brain processes temporal information.

A new insight into the filled duration illusion in the auditory modality: Difference between duration ranges

Emi Hasuo *Aoyama Gakuin University/Japan Society for the Promotion of Science, Japan*

The duration between the onset and the offset of a sustained sound is called a filled interval, and the duration between the onsets of two very brief sounds is called an empty interval. Filled intervals are often perceived to be longer than empty intervals, and this phenomenon is sometimes called the filled duration illusion. In this presentation, I will introduce our recent studies, which showed that the relationship between the perceived durations of filled and empty intervals can vary with duration range. When we measured the perceived durations of filled and empty time intervals of 30–600 ms using magnitude estimation, the perceived durations of filled intervals increased linearly as the interval lengthened, while for the empty intervals, the slope became gentler for intervals above about 100 ms than for shorter intervals. This may reflect a difference in the processing of empty intervals below and above this boundary.

Spatiotemporal interactions in perception and action: New approaches to the kappa effect

Tsuyoshi Kuroda *Shizuoka University, Japan*

The dependency of space and time in the cognitive system has recently attracted attention from neuroscience researchers. However, this issue was already a scientific concern early in the 20th century for researchers in the field of psychophysics. Indeed, a famous illusion, called the kappa effect, was first reported by a Japanese researcher in 1935. This effect occurs when two stimuli (or more) are presented successively. An empty time interval between the two stimuli is perceived as longer when these stimuli are located further away from each other in space. The effect is thus interpreted as indicating a perceptual tendency to overestimate duration when spatial distance is increased. I will introduce several studies that my colleagues and I conducted with this classical spatiotemporal illusion.

Neural correlates of auditory temporal assimilation: an EEG and MEG study

Takako Mitsudo *Kyushu University, Japan*

Auditory temporal assimilation is a unique perceptual phenomenon in which the duration of two successive empty intervals (T1 and T2) that are defined by three auditory stimuli are

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perceived as equal even when they are physically different. We investigated neuronal network dynamics of this phenomenon using electroencephalographic and magnetoencephalographic recordings. We observed activation in the right temporoparietal junction shortly after the second stimulus onset (i.e., the end of T1), which was stronger corresponding to the attention to T1. We also observed activation in the right inferior frontal gyrus at specific latencies after the end of T2, which was related to judging whether the two intervals had the same duration or not. The results provided a framework for spatiotemporal brain mechanisms that underlie the temporal judgment processes associated with attention and decision making that lead to the illusion.

Time Perception in Patients with Parkinsons Disease

Giovanna Mioni (1), Simon Grondin (2), Franca Stablum (1) *1. University of Padova, Italy; 2. Laval University, Canada*

It is known that Parkinson's patients (PD) show dysfunctions in time perception and that observing emotional facial expressions alter temporal judgments. In this conference, we review temporal dysfunction in PD patients and the magnitude of temporal distortions caused by the presentation of emotional facial expressions (i.e., happiness, anger, sadness and shame) in PD patients and controls. PD patients and healthy older adults were tested with a time bisection task with standard intervals lasting 400 ms and 1600 ms. PD patients were divided into two sub-groups, with and without mild cognitive impairment (MCI), based on their neuropsychological performance. The effect of facial emotional stimuli on time perception was evident and depended on the emotional facial expression presented. PD-MCI patients were more likely to underestimate long and overestimate short temporal intervals than PD-non-MCI patients and controls. Temporal dysfunction in PD-MCI patients is discussed in terms of memory dysfunction.

IS093
Sound and Time

Organizers: Sonoko Kuwano (1), Masanobu Miura (2) *1. Osaka University, Japan; 2. Ryukoku Uniersity, Japan*

Discussant: Minoru Tsuzaki Kyoto City University of Arts, Japan

Session Abstract:

Sounds convey information along the temporal stream. Therefore, the research into temporal factors is important in auditory perception. It is well known that basic attributes of auditory perception, i.e. loudness, pitch and timbre, are affected by temporal factors. Temporal aspects are also important in music and speech perception. Concerning applied research, when environmental problems are considered from psychological viewpoints, it can be made clear what are the important factors to create comfortable sound environment by examining the subjective impression along the temporal steam. In this symposium, basic psychophysics, music and speech perception, and environmental problems, including physiological approach, will be discussed focusing attention to temporal aspects.

Measuring loudness through reaction time: Binaural and temporal summation

Wolfgang Ellermeier (1), Josef Schlittenlacher (2) *1. Technische Universität Darmstadt, Germany; 2. Cambridge University, United Kingdom*

Auditory simple reaction time (RT) to the onset of short sounds is closely related to their loudness. This fact was further studied in two series of experiments investigating binaural summation and temporal integration of loudness. In study 1, listeners reacted to 200-ms tones covering a large range of levels, and either being presented binaurally, to their right, or to their left ear. RTs showed evidence for binaural loudness summation corresponding to a 6-dB gain, consistent with current loudness models. Study 2 further explored, which signal properties affect RT immediately after the onset of a sound, and how that relates to loudness. To that effect, rise time, duration, amplitude modulation frequency and phase of the signals were varied, all showing systematic effects on RT. To sum up: (Reaction) Time is a useful dependent variable to study auditory processing, as are the temporal parameters constituting independent variables when studying loudness.

The effect of quiet portion of intermittent noise on the overall impression

Makoto Morinaga (1), Hidebumi Tsukioka (1), Jiro Kaku (2), Sonoko Kuwano (3), Seiichiro Namba (3) *1. Defense Facilities Environment Improvement Association, Japan; 2. Kobayasi Institute of Physical Research, Japan; 3. Osaka University, Japan*

The effect of quiet portion of intermittent noise on the overall impression was examined by carrying out two psychological experiments. Intermittent noises composed of aircraft noise for event noise and soft road traffic/railway noise for ambient noise were used. In Experiment 1, the time interval between aircraft noises, i.e. the length of quiet portion, was varied by changing the number of aircraft noises and the total duration of stimuli was kept equal. On the other hand, the time interval between aircraft noises was varied by changing the total duration of stimuli and the number of aircraft noises was kept equal in Experiment 2. Participants were requested to judge the instantaneous and overall noisiness. The results suggested that the overall noisiness was almost decided by LAeq, in each stimulus. It was also suggested that overall noisiness tended to decrease according to the length of quiet portion if LAeq was equal.

Learning to hear a silence?: Behavioural and neurophysiological studies of Japanese specific speech listening

Makiko Sadakata (1), Sekiyama Kaoru (2) *1. University of Amsterdam / Radboud University Nijmegen, Netherlands; 2. Division of Cognitive Psychology, Kumamoto University, Kumamoto, Japan*

The way we perceive speech signals heavily relies on our prior linguistic experiences. This talk highlights a new example of such cases. The target speech signals are Japanese geminate consonants (e.g.,/ss/ and /kk/). In contrast to many languages that treat geminate consonants as a consonant cluster, Japanese phonetic and orthographic systems suggest segmenting them. For example, according to the Japanese systems, a pseudo word "assu" should be segmeted into three morae units, /a//Q//su/, where /Q/ is a phonetic representation of a moraic voiceless obstruent. While /Q/ has often been described based on speech production data, not much is known whether Japanese native speakers rely on /Q/ when perceiving geminate consonants. Our behavioural and neurophysiological experiments suggested that Japanese native speakers seem to rely on /Q/ in perception, that they represent /Q/ as somewhat similar to a silent duration, and that such a representation seem to be reflected in neurophysiological signatures.

Embodiment of metrical structure: motor patterns and two fMRI studies

Chen-Gia Tsai (1), Li-Ching Wang (2) *1. National Taiwan University, Graduate Institute of Musicology, Taiwan; 2. University of Cambridge, Centre for Music and Science, United Kingdom*

Sensory feedback, whether auditory, visual, tactile, proprioceptive and vestibular, enables music performers to perceive metrical structures of music better due to the multiple sources of information. Cognitively, humans tend to synchronize their body movements with beats they are listening to. The present show that embodiment of metrical hierarchy can also be observed in the variety of beat-counting processes from different musical cultures. In addition, we used fMRI to demonstrate how the activation pattern of the sensorimotor network was modulated by the sequential nature of auditory input

IS094
The many faces of face perception

Organizer: Yumiko Otsuka *Ehime University, Japan*

Session Abstract:

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Faces are rich visual stimuli that provide a diversity of information in one harmonious whole. From a glance at someone's face we can perceive their identity, sex, age, attractiveness, and attentional and emotional state and many other of their characteristics. These perceptions shape our subsequent interactions. This symposium will illustrate some of this diversity of visual information processing by bringing together speakers looking at a number of different aspects of face perception including 1) face processing in autism spectrum disorder, 2) the nature of biases in estimating others' sex and age from faces, 3) how we determine others' gaze direction, 4) what the hollow face illusion reveals about three dimensional shape perception and 5) illusory effects of facial makeup and cosmetics. We expect the symposium to be relevant to a wide audience with an interest in perception in general and faces in particular.

The dual-route model of the influence of head orientation on eye gaze perception

Yumiko Otsuka (1), Isabelle Mareschal (2), Colin Clifford (3) *1. Ehime University, Japan; 2. Queen Mary University of London, United Kingdom; 3. UNSW Australia, Australia*

It has long been known that perceived gaze direction is not determined solely by the information given in the eyes, but also influenced by head orientation. Despite numerous previous studies, it has been unclear how the perceived gaze direction is determined by these two cues. This talk will describe the dual-route model (Otsuka et al., 2014, 2015) we proposed recently, which computes perceived gaze direction as a linear combination of eye deviation and head orientation. The dual-route model clarifies the conditions that reveal previously identified attractive and repulsive influences of head orientation, and explains the well-known Wollaston effect as the misapplication of a process that normally maintains gaze constancy in spite of changes in the visible part of the eyes with head rotation. Further, we show that a simple linear combination of eye deviation and head orientation is sufficient to explain the perceived gaze direction.

Expectation and perception of the age and sex of a face

Tamara L Watson (1), Yumiko Otsuka (2), Colin W Clifford (3) *1. Western Sydney University, Australia; 2. Ehime University, Japan; 3. UNSW Australia, Australia*

Prior expectation may influence our perception of people at the first meeting. For example, it has been shown that we are more likely to categorise a person as 'male' or to estimate their age as closer to our own. This could be due to perceptual effects and/or cognitive response bias. We use a Bayesian modelling approach to estimate perceptual biases inherent in our decisions about the sex and age of faces. Under conditions of increasing uncertainty participants are more likely to respond 'male' and 'older'. This is consistent with the operation of a Gaussian prior with a peak towards the male and older faces and with the prior becoming more inflectional under conditions of increased uncertainty. This demonstrates a biased expectation, operating at a perceptual level, that faces will be male and older. We anticipate that it represents a true inaccuracy in the experienced percept under conditions of uncertainty.

Impaired face processing in Autism Spectrum Disorders (ASD) may arise from an early bias for objects.

Karen R Dobkins (1), Leslie J Carver (1), Natacha Akshoomoff (1), Pam Pallet (2), Joseph Mccleery (3) *1. University of California, San Diego, United States of America; 2. The Ohio State University, United States of America; 3. Children's Hospital of Philadelphia, United States of America*

Previous studies have reported face processing impairments in individuals with Autism Spectrum Disorders (ASD), however, the origins remain unknown. To address this question, we studied face processing, in comparison to object processing, in adolescents diagnosed with ASD, as well as in infants who are "High-Risk" for developing ASD (because they have an older sibling with ASD). In line with previous studies, our perceptual studies of adolescents with ASD reveal impaired performance for discriminating faces, however, this effect is relatively small compared to the finding of enhanced discrimination of objects (cars). Our findings in adolescents are mirrored in neural data (obtained with event-related potentials) from "High-Risk" infants. At 10 months, these infants show relatively normal response to faces, yet enhanced responses to objects (toys). Together, these results suggest that impaired face processing in ASD may arise from an early bias for objects, which hinders the typical development of face processing.

Illusory effects of facial makeup

Kazunori Morikawa *Osaka University, Japan*

Does eye makeup actually make the eyes appear larger than they really are? If so, by what percentage? To answer these questions, we measured perceived eye size with 4 levels of eyeliner and 5 levels of mascara, using a psychophysical method. The results showed that both eyeliner and mascara make the eyes appear larger by up to 6% (13% in area), but their effects are not additive. We also measured perceived eye size with eye shadow and demonstrated that eye shadow increases perceived eye size by about 5% (10% in area). These findings indicate that facial makeup can induce visual illusions. In addition, it is suggested that the eye size illusion caused by eyeliner, mascara, and eye shadow employs the same mechanism as that of the Delboeuf illusion, a geometric illusion of assimilation. Moreover, we found that the position of the eyebrows influences the perceived shape of the eyes.

The hollow-face illusion and three-dimensional shape perception

Harold C Hill *University of Wollongong, Australia*

The hollow-face illusion, the perception of a concave mask as a convex face, suggests that face perception goes beyond the two dimensional arrangement of features. It can also help us understand how the perception of three-dimensional face shape is achieved, providing a challenging test case for models of depth cue integration more generally. Results will be reviewed on the effects of lighting, binocular disparities, familiar orientation, surface reflectance, refractive error, and depth-of-field on two measures of the strength of the illusion: flipping distance and flipping contrast. Recent work shows that the ocular cues of convergence and accommodation are driven by illusory depth even when disparity, blur and contrast based error signals are available but that ocular information may be critical in disambiguating the illusion at short viewing distances. Implications for matching faces across changes in viewpoint and lighting, and for the diagnostic value of the illusion clinically, will be considered.

CONTRIBUTED SYMPOSIUM

CS112
Symposium on the Perceptual Basis of Aperture Passability Judgment

Organizers: Nobuhiro Furuyama (1), Hiroyuki Mishima (1), Shin Maruyama (2), Tetsushi Nonaka (3) *1. Waseda University, Japan; 2. Komazawa Women's University, Japan; 3. Kobe University, Japan*

Session Abstract:

The issue of whether and how we can perceivepassability of apertures is one of the most important topics of study on affordance perception (Warren & Whang, 1987; Gibson, 1966, 1979). One of the aims of the present symposium is to lay out some (if not all) of the recent studies on our ability to make such judgments about a given aperture by various means of locomotion (walking, running, etc.), with or without an external object attached to the body (a-person-sitting-in-a-wheel-chair, a-person-walking-with-a-handheld-object, a-person-walking-and-pushing-a-cart, etc.), and in a variety of situations (single or multiple apertures). Despite the accumulating empirical studies, however, there has not been a single definitive theory that explains all the data at this moment. The other aim of this

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symposium, then, is to discuss the perceptual basis of the passability judgment, and what the data presented in this symposium reveal about these theories.

Perception of dynamically evolving affordances for passing through apertures

Jeffrey B Wagman *Illinois State University, United States of America*

Safely and efficiently performing a given behavior requires perceiving whether that behavior is possible, and if so, how the body (and objects attached to it) must be controlled in doing so. Such possibilities for behavior that depend on the relationship between action capabilities and the features of a given situation are known as affordances. Importantly, typical environments are not only highly cluttered but the features of those environments are continually changing. Moreover, the action capabilities of a given perceiver are also continually changing. Therefore, perceiving dynamically evolving affordances for passing around or between obstacles is critical for the everyday behaviors performed by humans and nonhumans alike. This talk will review research investigating perception of affordances for passing through apertures by humans and by other species. I conclude that perceiving affordances for any behavior (including passing through apertures) by any species (including humans) requires assembling a task-specific smart perceptual instrument.

Visuomotor control of locomotion through apertures: how body-scaled information is obtained

Takahiro Higuchi *Tokyo Mertopolitan University, Japan*

Individuals generally rotate their body when an opening is narrower than 1.1-1.3 times their shoulder width. This rotation is initiated generally two steps prior to entering the opening and its amplitude is determined so that it produces a minimum spatial margin under safe situations. The prerequisite of such behavior is that individuals can perceive "the width of an opening relative to the body width," or more generally, "the environmental properties relative to one's action capabilities" very accurately when the opening is still far from them. Such information is often referred to as body-scaled (or action-scaled) information. The purpose of my presentation in this symposium is to show how such information is obtained while approaching toward an aperture.

The body-pushcart coordination in aperture crossing when there are more than one aperture

Takayuki Tomono *Waseda University, Japan*

Studies on affordance perception for aperture crossing (Higuchi et al., 2004; Higuchi et al., 2011; Wagman and Taylor, 2005; Warren and Whang, 1987) have elegantly shown that it is based on an intrinsic body-scaled ratio (e.g., a ratio of aperture width to actor's shoulder width), referred to as the pi number (Warren, 1984). We live in the cluttered environment, however, and are likely to encounter situations where there are more than one aperture. In this study, the participants were instructed to choose the narrowest aperture that they were able to pass through with a pushcart of different sizes from among the four apertures set up for the experiment, and to actually pass it through with a pushcart. The results suggest that the participants coordinated their body with a pushcart before and while passing through the chosen aperture.

CS113
The Contents of Perception and the Perception/Cognition Border.

Organizers: Eric Mandelbaum, Eric Mandelbaum *City University of New York (Baruch), United States of America*

Session Abstract:

Our symposium addresses the question of perceptual content, in particular the range of properties that perception produces without the aid of cognition, and how culture and experience change these outputs. We offer a variety of perspectives on these issues. Tim Bayne (Western Ontario) argues for the liberal view that perceptual content includes high-level representational properties, such as scene gist and ensemble properties, contents that are affected by the culture in which one finds oneself. Taking a more conservative view, Eric Mandelbaum (CUNY) presents data that suggest that perception categorizes only basic-level and evolutionarily significant properties (e.g., causation, agency) without the aid of cognition. Ian Phillips (Oxford) offers evidence that the data used to underwrite arguments about perceptual and phenomenal content are in fact neutral on the question of content, and concludes that we still don't know how to move beyond introspection in approaching the question of content.

Seeing and Conceptualizing: Modularity and the Shallow Contents of Perception

Eric Mandelbaum *City University of New York (Baruch), United States of America*

This paper investigates the relation between categorization, a perception/cognition border, and the contents of perception. After presenting evidence about categorization behavior the paper argues for the following theses: 1) that there is a border between perception and cognition; 2) that the border is to be characterized by perception being modular (and cognition not being so); 3) that perception outputs conceptualized representations, so views that posit that the output of perception as solely non-conceptual are false; and 4) that perceptual content consists of basic-level categories and not richer contents.

Ensemble Encoding and the Contents of Perception

Tim Bayne (1,2) *1. University of Western Ontario, Canada; 2. University of Manchester, United Kingdom*

This talk examines the bearing of ensemble encoding on the debate about where the border between perception and cognition is to be drawn and whether perception can represent 'high-level' properties. The kind of ensemble encoding on which this paper focuses concerns the representation of mean properties of the presented array. Humans have been shown to be extremely reliable at detecting the mean for a wide range of properties, including: orientation, location, direction of gaze, direction of motion, facial expression of emotion, gender and racial identity. Does the representation of these properties fall on the perception side of the perception/judgment divide, or does it fall on the judgement side? And if ensemble encoding is perceptual, what bearing does that have on debates concerning the admissible contents of perception? This paper argues that ensemble encoding is genuinely perceptual, and that it provides evidence that 'high-level' properties can be perceptually represented.

No watershed for overflow: recent work on the richness of consciousness

Ian B Phillips *University of Oxford, United Kingdom*

Does phenomenal experience "overflow" availability to cognition? Do we consciously see more than we can remember and report? Psychologists and philosophers have long sought to move beyond naïve appeals to introspection by providing empirical evidence for or against overflow. In particular, recent data from change detection paradigms involving the representation of statistical features or completed surfaces have been interpreted as powerful evidence in favour of overflow. I explain why the relevant data are quite consistent with a "no overflow" interpretation. I further argue that there is no difference between "overflow" and "no overflow" interpretations when they are framed purely in representational terms. A difference only emerges when additional and tendentious assumptions are made concerning which representations correspond to elements in conscious experience. A sobering moral is drawn: we still do not know how to move beyond appeals to naïve introspection in establishing the nature and limits of our ordinary experience.

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Seeing and Conceptualizing: Modularity and the Shallow Contents of Perception

Eric Mandelbaum *City University of New York (Baruch), United States of America*

This paper investigates the relation between categorization, a perception/cognition border, and the contents of perception. After presenting evidence about categorization behavior the paper argues for the following theses: 1) that there is a border between perception and cognition; 2) that the border is to be characterized by perception being modular (and cognition not being so); 3) that perception outputs conceptualized representations, so views that posit that the output of perception as solely non-conceptual are false; and 4) that perceptual content consists of basic-level categories and not richer contents.

CS114
Consciousness is ever-present, but determined by the past: How the perceptual past influences the perceptual now

Organizer: Ikuya Murakami *The University of Tokyo, Japan*

Session Abstract:

There are many examples of how conscious visual perception is affected by the history of previous stimulus presentation. These effects are sometimes negative, in that they lead to the opposite percept to the one perceived, while others are positive, facilitating the processing of a repeated stimulus. Examples of low-level adaptation are aftereffects such as motion and color afterimages which are both negative. Secondly, there are mid-level history effects in perception such as from face adaptation, which also are negative. Thirdly, higher-level attention-based history effects have been shown to have a dominating influence upon attention shifts, but these differ from the former in that they facilitate the perception of the same stimulus. The ultimate aim of the symposium will be to try to identify ways in which theoretical accounts of one and the other can mutually inform each other, for a unified conception of how past perception determines present perception.

The Continuity Field (CF): a mechanism for perceptual stability via serial dependence

David Whitney *UC Berkeley, United States of America*

A critical function of vision is to stabilize perception, so objects look the same from moment to moment. This is a challenge because visual input is noisy and discontinuous. Though a classic question, the mechanism that links the perceived identity of an object from moment to moment is unknown. We recently discovered the Continuity Field (CF), a mechanism of object constancy built on serial dependence; the present appearance of an object is captured by what was perceived over the last several seconds. The spatiotemporal tuning of this sequential dependence reveals the continuity field, within which perceptual judgments are dictated by previous percepts--making different objects appear the same. The continuity field is a physiologically plausible operator that is not due to memory or hysteresis, it affects perception and not just decisions or motor responses, and it reveals a novel mechanism by which the visual system achieves stable object representations over time.

Utility of prior exposure for perceivers, for searchers, and for researchers

Ikuya Murakami (1), Masahiko Terao (1,2), Takayuki Osugi (1,3) *1. The University of Tokyo, Japan; 2. Yamaguchi University, Japan; 3. Japan Society for the Promotion of Science, Japan*

Prior exposure to visual stimuli can alter subsequent visual processing. Adaptation to a particular stimulus often induces a negative aftereffect in a subsequently viewed stimulus at the same retinal position. An experience of a priming stimulus tends to bias the perceptual and attentional sets of the observer either toward or away from it. Here I will argue the usefulness of such prior exposure from the viewpoints of an observer conducting a detection task or a visual search task and of a researcher seeking a mental representation related to some visual processing that is hardly accessible in other ways. My argument involves psychophysical findings such as improved motion-direction discrimination after adaptation to a task-irrelevant orthogonal motion, improved attentional deployment after exposure to visual instruction about where not to search, and taking advantage of visual adaptation to investigate surface representation in the visual system.

Are perceptual and attentional memory two sides of the same coin?

Arni Kristjansson *University of Iceland, Iceland*

Both attentional selection in visual search and perceptual selection in bistable perception are sensitive to trial history: They are biased toward previously selected targets or interpretations, and they operate on similar timescales. We tested their mutual interactions, presenting either ambiguous spheres that could be perceived as rotating in one of two directions, or search displays in which unambiguously rotating target and distractor spheres closely resembled the two possible interpretations of the ambiguous stimulus. We interleaved both trial types within experiments, testing whether priming by target selection during search would affect the outcome of bistable perception and, conversely, whether sensory memory during bistable perception would affect search times. While we found intertrial repetition effects among consecutive search trials and among consecutive bistable trials, there were no cross-paradigm effects. This was surprising given demonstrations of similarity but recent proposal that sensory memory is tightly linked to perceptual interpretation may explain this independence.

CS115
Perception of material properties

Organizer: Shin'ya Nishida *NTT Communication Science Labs, Japan*

Session Abstract:

Past extensive research on visual perception provide us with reasonable understanding as to how we perceive basic visual attributes, such as color, shape, motion and depth. Mere combinations of such basic attributes however cannot fully explain our perceptual experiences under real natural environments. Just by looking, we are able to estimate the objects' material properties (or Shitsukan in Japanese). Speaking more technically, by decoding complex high-dimensional information embedded in the image of the objects, our visual system is able to estimate the object's optical reflectance property (e.g., gloss), mechanical property (e.g., viscosity), physical condition (e.g., wet) and material category (e.g., metal). This symposium will introduce recent advancements in psychophysics about this marvellous ability of the human visual system, featuring three talks by Karl Gegenfurtner (Giessen University, Germany), Isamu Motoyoshi (University of Tokyo, Japan) and Shin'ya Nishida (NTT Communication Science Labs, Japan).

The mental and neuronal representation of material properties

Karl R Gegenfurtner *Giessen University, Germany*

The past 15 years have seen an increased interest in the perception of materials and their properties. I will present findings showing that the classification of images into different material categories is as fast as an analogous task with object classes. Using linear classifiers, material categories could be differentiated based on the ERP pattern in single trials as early as 100 ms after stimulus onset. In a series of experiments with sighted and blind observers, we found that there is a systematic perceptual representation of materials that transcends the visual and haptic modalities. Neuroimaging experiments showed that perceptual judgments on material properties coincided quite well with an analysis of low level image features and with the activity in early visual cortical areas. Overall, there has been a remarkable advance in our knowledge of the perception of materials and material categories, at the neuronal, perceptual and computational levels.

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Perceiving the quality of surfaces and images

Isamu Motoyoshi *The University of Tokyo, Japan*

Recent psychophysical studies reveal visual information processing underlying our astonishing ability of perceiving physical properties and material of natural surfaces. These studies now lead us to investigate visual mechanisms that support the perception and assessment of various kinds of 'qualities' and 'impressions' in the image ('Shitsukan'). In this talk, I will introduce several studies carried out along this direction, including the perception of different types of blur in natural photographs, and the human preference for particular surfaces and images. I will also discuss impact and limitation of low-level image features in these quality judgments.

Visual recognition of Shitsukan

Shin'ya Nishida *NTT Communication Science Labs, Japan*

Recognition of Shitsukan is a marvellous ability of human's brain to read out the nature of objects. We obtain indispensable information for our living from Shitsukan sensed through multiple modalities. Shitsukan plays important roles not only in perception, but also in value-based behaviour selection and body motor control. How our brain reads out Shitsukan from complex high-dimensional source information embedded in sensory signals is a hard but critical research question for scientific understanding of human sensory processing and advancement of information technologies. In my talk, I will review recent studies on Shitsukan perception mainly carried out in our lab, including a computational theory to estimate surface reflectance properties in parallel with shape estimation, visual perception of liquid viscosity, scene wetness and texture fineness, integration of Shitsukan information across sensory modalities, and manipulation of visual appearance of real objects by projection mapping.

CS116
Visual memory, statistical processing and attention in understanding visual environments

Organizer: Jun Saiki *Kyoto University, Japan*

Session Abstract:

Visual cognition research has shown that our understanding of visual environments, including objects, scenes and events, emerges through interaction of visual memory, statistical processing and attention. Although previous studies revealed cognitive and neural mechanisms underlying these functions, many questions remain unsolved. In everyday situations, interactions between statistical processing in extracting gist and layout of a scene, and visual working memory in maintaining object representations are essential, but statistical processing and visual working memory have been studied separately. For the interaction between statistical processing and visual working memory, visual attention should play a critical role in their coordination, and empirical studies are needed. To facilitate our understanding of visual cognition in complex environments, this symposium features three experts in this field reporting their recent findings on statistical processing, visual working memory and attention, and discusses future directions of visual cognition research.

The effect of background image statistics on perceived object size

Chien-Chung Chen *National Taiwan University, Taiwan*

The perceived size of an object, as demonstrated in the Ebbinghaus illusion, depends on the size of the context elements. We further demonstrated that it is also influenced by the image statistics of a non-uniform context. We measured the perceived size of a target embedded in various backgrounds. The background texture consisted of 5000 randomly distributed disks whose size drawn from distributions varied in mean (60 to 600 arcmin), standard deviation (0 to 0.27 fold of the mean) and skewness (−0.37 to 0.37). The perceived target size decreased with the mean size of the background elements to a critical value and then showed no further change with background. The variance of the background element size did not affect the perceived target size. The negatively skewed background decreased perceived target size for small backgrounds but increased for large backgrounds. Our result suggests a size normalization process underlying size perception.

Left-side bias on statistical information extraction

Su-Ling Yeh, Kuei-An Li *National Taiwan University, Taiwan*

Our understanding of how we perceive visual environments have improved from previous visual cognition research, but many questions remain debated or uninvestigated. In particular, there is a phenomenon called "left-side bias" (LSB) where perceptual judgments tend to bias toward the information from the left visual field. We are interested in how LSB affects statistical information extraction from visual input such as mean size. While contemporary studies have ascertained this difference in perceptual weighting from both sides of visual field, they still lack information to predict exactly how the left side is represented compared to the right side. In order to answer that question, we conducted experiments using mean size adjustment tasks. We found that LSB is induced by a leftward attentional bias and that LSB magnifies the weighting of mean size from the left side.

Neural Mechanisms by Which Attention Modulates the Comparison of Remembered and Perceptual Representations

Bo-Cheng Kuo *National Taiwan University, Taiwan*

Attention is important for effectively comparing incoming perceptual information with the contents of visual short-term memory (VSTM). However, how attentional mechanisms operate upon these comparison processes remains largely unknown. Here we investigate the underlying neural mechanisms by which attention modulates the comparisons between VSTM and perceptual representations using functional magnetic resonance imaging (fMRI). Participants performed a cued change detection task. Spatial cues were presented to orient their attention either to the location of an item in VSTM prior to its comparison (retro-cues), or simultaneously with the probe array (post-cues). When attention cannot be effectively deployed in advance (e.g. post-cues), we observed distributed prefrontal and parietal activations in support of successful change detection. Moreover, we demonstrated a close relationship between this prefrontal-parietal functional connectivity and behavioral performance. Our findings elucidate the attentional and neural mechanisms by which items held in VSTM are compared with incoming perceptual information.

Binding of non-spatial features in visual working memory

Jun Saiki *Kyoto University, Japan*

The mechanism of binding non-spatial features in visual working memory remains unknown. In the current study, I modified a redundancy-gain paradigm to investigate this issue. A set of features was presented in a two-object memory display, followed by a single object probe. Participants judged whether the probe contained any features of the memory display, regardless of its location. Response time distributions revealed feature coactivation only when both features of a single object in the memory display appeared together in the probe, regardless of whether the probe and memory objects share the same location. This finding suggests that a shared location is necessary in the formation of bound representations but unnecessary in their maintenance. Electroencephalography data showed that amplitude modulations reflecting location-unbound feature coactivation were different from those reflecting the location-sharing benefit, consistent with the behavioral finding that feature-location binding is unnecessary in the maintenance of color-shape binding.

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THEMATIC SESSION

TS109
The visual phantom illusion revisited

Organizer: Jiro Gyoba *Tohoku University, Japan*

Session Abstract:

The visual phantom illusion is a kind of perceptual completion phenomenon, in which the partly occluded grating appears to continue across the occluder. Since the late 1980s, a long series of psychophysical experiments have been conducted by the researchers of this session, in close relation to the grating induction effect, a strong brightness contrast illusion, discovered by M.E. McCourt. Based on recently found perceptual phenomena and experimental results, we discuss visual phantoms as a phenomenon of surface completion which produces the modification of early visual processing dependent on perceptual organization. As an evidence suggesting the surface completion, a compelling new illusion (Phantom Induced Blindness, PIB) is introduced in which dots placed on the stationary inducing gratings perceptually disappear when the phantom gratings are perceived. Three vision scientists will introduce novel phenomena and data differentiating the visual phantom phenomenon from the grating induction effect.

Phantom induced blindness

Jiro Gyoba *Tohoku University, Japan*

In order to indicate visual phantoms as a phenomenon of surface completion, we discovered a new compelling illusion, phantom induced blindness (PIB) in which high intensity yellow dots placed on the stationary inducing gratings perceptually disappear when the phantom gratings are perceived. When the luminance of the occluder was near to the average luminance of the inducing gratings, the grating induction effect (GI) became dominant, resulting in much shorter duration of the dot disappearance. We also found the magnitude of PIB decreased under the inducing grating with higher spatial frequency that made the visibility of phantoms weaker. These results suggested the surface completed by visual phantoms could hide the dots and produce perceptual blindness.

Visual phantoms induced by second-order components

Kenzo Sakurai *Tohoku Gakuin University, Japan*

It has been a long controversial issue whether the visual phantom illusion solely depends on the low-level brightness detection mechanisms. When an occluder has a mean luminance of inducing gratings, visual phantoms disappear and an out-of-phase grating induction effect takes over. Second-order components of inducing gratings, however, produce visual phantoms even when the occluder has a mean luminance of inducing gratings. This suggests that the grating induction effect interfere with the perception of the phantoms by first-order components, especially when the occluder has a mean luminance of the inducing gratings. With some demonstrations of various kinds of second-order component (texture, color, contrast-modulation, etc), I will show that both moving and stationary visual phantoms induced by second-order components are robust against the luminance change of the occluder, and that the grating induction effect interfere with the visual phantoms when both are induced by first-order components.

Visual phantoms and perceptual transparency

Akiyoshi Kitaoka *Ritsumeikan University, Japan*

The visual phantom illusion is explained with not only brightness induction but also perceptual transparency (Kitaoka, Gyoba and Sakurai, 2006). If induced brightness in the occluder is in phase of the inducing grating, bistable transparency works. This type of phenomenon is observed in 'photopic' phantoms (Kitaoka, Gyoba and Kawabata, 1999) or possibly in moving visual phantoms. If induced brightness in the occluder is counterphase of the inducing grating, unique transparency governs. This type of phenomenon is embedded in stationary phantoms (Gyoba, 1983; Gyoba and Sakurai, 1985) that shares the same brightness induction effect as the grating induction (McCourt, 1982). Further research on visual phantoms has suggested a close relationship to some brightness/lightness illusions that are supposed to be based upon figure-ground segregation, e.g. the Anderson illusion (Anderson and Winawer, 2005).

TS110
Grapheme-color synesthesia in East Asian languages

Organizers: Kazuhiko Yokosawa (1), Michiko Asano (2) *1. The University of Tokyo, Japan; 2. Rikkyo University, Japan*

Session Abstract:

Grapheme-color synesthesia is a condition in which a grapheme (a letter, or a character) induces a specific color sensation, which affects a small percentage of the population. Recently, research on grapheme-color synesthesia has revealed certain regularities, as well as individual differences in grapheme-color associations. Such regularities could inform us of underlying mechanisms of grapheme-color synesthesia. Many studies have shown that basic linguistic features, such as grapheme frequency, sounds, and meaning influence synesthetic color choices. Moreover, studies on grapheme-color synesthesia in East Asian languages, which have distinctive features that are absent in English, have made unique contributions to this research. This symposium aims to present and discuss recent findings on grapheme-color synesthesia in East Asian languages, including Japanese and Korean. In the symposium, findings from a variety of perspectives will be discussed, including language processing, development, cross-linguistic investigation, perceptual property of synesthetic colors, and grapheme-color synesthetic tendency in non-synesthetic populations.

Synesthesia-like associations between graphemes and colors in Japanese non-synesthetic population

Junichi Nagai (1), Kazuhiko Yokosawa (2), Michiko Asano (3) *1. University of the Sacred Heart, Japan; 2. The University of Tokyo, Japan; 3. Rikkyo University, Japan*

We examined associations between graphemes and colors in Japanese non-synesthetic population. Participants chose the most suitable color from 11 basic color terms for dozens of graphemes respectively, that are used in the Japanese language. In their responses, participants showed biases and regularities that were comparable to those of synesthetes reported in past studies. Although it has been believed that only synesthetes, and not non-synesthetes, tended to associate graphemes with colors based on grapheme frequency, Berlin and Kay's color typology, and color word frequency, participants in our study tended in part to associate graphemes with colors based on above factors. Moreover, our participants tended to associate different graphemes that shared sounds and/or meanings with the same colors, which was analogous to the findings in Japanese synesthetes. These results support the view that grapheme-color synesthesia might have its origins in cross-modal association processes that are shared with the general population.

Developmental Changes of Synesthesia Observed in Japanese Elementary School Children

Eiko Matsuda (1), Okazaki Yoshihiro (2), Froese Tom (3) *1. Research Center for Advanced Science and Technology, The University of Tokyo, Japan; 2. Graduate School of Education, Okayama University, Japan; 3. Departamento de Ciencias de la Computación Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas Universidad Nacional Autónoma de México, Mexico*

This study focuses on the prevalence of number-color synesthesia and number personification (Simner and Holenstein, 2007), in children attending Japanese elementary schools. In our previous study, we found surprisingly high prevalence of number personification in Japanese society (Matsuda et al., 2013). We therefore became interested in studying the

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process to form this type through development. We conducted a paper-based questionnaire with 211 participants, asked them to describe the colors and personalities of the numbers 0–9, and computed consistency of their reports as an index of synesthesia. The results showed that the number personification score decreased between the 4th and 6th grade children (9 to 12 years old), while that of number-color synesthesia remained the same. Between these ages, children go through a developmental transition from the stage of concrete operation to that of formal operation (Piaget, 1954). Finally, we discuss the relationship between synesthesia and children's formal operation ability.

Determinants of synesthetic colors for different types of graphemes, Japanese characters and the English alphabet: a developmental model

Michiko Asano (1), Kazuhiko Yokosawa (2) *1. Rikkyo University, Japan; 2. The University of Tokyo, Japan*

Recently, grapheme-color synesthesia research has revealed that certain regularities, as well as individual differences, figure into grapheme-color associations. Although several factors are known to regulate grapheme-color associations, the impact of factors, including their interrelationships, on synesthesia remains unclear. We investigated determinants of synesthetic color for graphemes of Hiragana, a phonetic script in the Japanese language, and the English alphabet. Results revealed that grapheme ordinality was the strongest predictor of synesthetic colors for Hiragana characters, followed by character sound, and visual shape. Ordinality and visual shapes also significantly predicted synesthetic colors for English alphabet letters, however, sounds did not. The relative impact of grapheme properties on grapheme-color associations and the differences between these two writing systems are accounted for by considering the way graphemes are processed in the brain and introduced during an individual's development. A new model is proposed which takes into account the developmental process of grapheme learning.

Non-random association between sound and color- Insight from multi-lingual grapheme-color synesthetes and non-synesthetes

Chai-Youn Kim, Mi-Jeong Kang, Ye-Seul Kim, Hyun-Woong Kim *Korea University, Republic of Korea*

Individuals with grapheme-color synesthesia experience "colors" when viewing achromatic letters and digits. Despite the large individual difference in synesthetic association between inducing graphemes and induced colors, graphemes sharing a certain perceptual or linguistic feature tend to be associated with similar synesthetic color. This talk is about a series of studies suggesting the importance of sound determining synesthetic color. The first study with a cross-linguistic approach shows that graphemes with similar sounds tend to induce analogous synesthetic colors. The second study examines whether graphemes within a same phonetic category across Korean hangul, Japanese hiragana and katakana as well as Latin alphabets tend to induce similar synesthetic colors. The last study further explores whether phonetic properties are associated with colors in a specific manner with other visual and linguistic features of graphemes removed. These results imply non-random association between sound and color not only in synesthesia but also in more general context.

Exploratory analysis of synesthetic colors distribution in the color space

Daisuke Hamada, Hiroki Yamamoto, Jun Saiki *Kyoto University, Japan*

This study investigated distribution of synesthetic colors in the color space to understand perceptual properties of synesthetic colors. This analysis requires a large amount of data for a single synesthete. We therefore collected large samples of data from five synesthetes. We obtained over 1000 synesthetic colors associated with Japanese kanji characters for each of two synesthetes, and over 100 synesthetic colors for each of the other synesthetes. We then analyzed the distribution pattern of synesthetic colors in the color space by using techniques of spatial statistics. The spatial statistical analysis revealed that synesthetic colors are concentrated in multiple regions in the color space, that is, they form "synesthetic color clusters". The synesthetic color clusters indicate that synesthetic colors are not selected randomly from the color space, but specific colors preferentially become synesthetic colors for each synesthete.

ORAL PRESENTATION

OR1829

Cultural Diversity in Eye Movements is shaped by Nurture not Nature

Roberto Caldara, Anne-Raphaelle Richoz, Yingdi Liu, Junpeng Lao *University of Fribourg, Switzerland*

Humans adapt to their environment through a unique amalgamation of culture and biology. Historically, it has long been presumed that across cultures, all humans perceive and sample the world in a comparable manner. Recently, however, this notion has been disputed by evidence showing fundamental perceptual differences between people from Eastern and Western cultures, for low- (i.e., reflexive saccades) and high-level (i.e., faces) visual information sampling strategies. Yet, whether these well-established cultural differences are related to nature or nurture remains unaddressed. We recorded the eye movements of culturally Westerners South Korean adoptees while they performed an old/new face recognition task and visually guided saccades. A Bayesian model assessed that the South Korean adoptees perfectly fitted with the Western oculomotor behavior for face recognition, and express reflexive saccades. Altogether, our data show that nurture and not nature is responsible for cultural variations in eye movements and visual information sampling.

OR1830

Visual functions improvement in mild myopia and amblyopia with combined perceptual learning and transcranial random noise stimulation

Gianluca Campana (1), Rebecca Camilleri (1), Andrea Pavan (2), Antonella Veronese (3), Giuseppe Lo Giudice (1,3) *1. University of Padova, Italy; 2. University of Lincoln, United Kingdom; 3. San Antonio Hospital, Padova, Italy*

During the last decade it has been shown how perceptual learning, an improvement in a sensory/perceptual task following practice, can generalize and transfer to unrelated visual functions such as visual acuity (VA) in participants with either refractive (e.g. myopia) or cortical deficits (e.g. amblyopia). It has recently been proposed that perceptual learning can be boosted by high-frequency transcranial random noise stimulation (hf-tRNS). By using different groups of participants (single-blind study), here we tested the efficacy of short contrast-detection trainings (8 sessions) with concurrent hf-tRNS, in improving VA and contrast sensitivity (CS) of adults with uncorrected mild myopia or with amblyopia. Results show that a short contrast-detection training is able to improve VA and CS both in participants with mild myopia or with amblyopia only if coupled with hf-tRNS, whereas no effect were seen with the sole administration of perceptual learning or hf-tRNS.

OR1831

Perceived duration of piano sounds

Emi Hasuo (1), Yoshitaka Nakajima (2) *1. Aoyama Gakuin University/Japan Society for the Promotion of Science, Japan; 2. Kyushu University, Japan*

Sounds in our everyday life have complex temporal structures, while the sounds in perceptual experiments are often physically simple with steady amplitude. We focused on the sound of the piano as a common musical instrument, and examined whether the perceived duration of a piano sound, which decays after rapidly

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reaching maximum amplitude, differs from that of a steady-state sound, which has constant amplitude. Perceived durations of (1) synthesized sound with decaying amplitude, (2) synthesized sound with steady amplitude, (3) recorded piano sound with decaying amplitude, and (4) recorded piano sound with very gradual decay in amplitude (i.e., piano sound recorded and processed to imitate the sound with steady amplitude) were measured for two physical durations (110 and 280 ms) using magnitude estimation. Decaying sounds of these physical durations were perceived to be shorter than sounds with steady amplitude for both synthesized sounds and piano sounds.

OR1832
Adaptation to Illusory Thermal Uniformity Induced by Thermal Referral

Hsin-Ni Ho (1), Doris Chow (1), Warrick Roseboom (1,2) *1. NTT Communication Science Laboratories, Japan; 2. University of Sussex, United Kingdom*

When thermal stimulators are touched with the middle three fingers of one hand but only the outer two fingers are heated, illusory warmth is perceived on the middle finger. This phenomenon, known as thermal referral (TR), produces an effect such that all three fingers feel a uniform thermal sensation. To clarify the level at which TR occurs we examined how adaptation to TR affects subsequent thermal perception. When subjects reported about all three fingers together, or the middle finger in isolation (non-stimulated), we found that the effect of TR adaptation was similar to having adapted to the subjectively matched physically uniform intensity. However, when reporting the index finger in isolation (directly-stimulated), the TR adaptation effect was similar to having adapted to the presented physical rather than illusory intensity. These results indicate that TR is not caused only by low-level thermoceptive pathway operations, but also involves computation at cortical level.

OR1833
Attention influence on binaural interaction in human auditory brainstem response under random distribution of sound locations

Kazunari Ikeda *Tokyo Gakugei University, Japan*

Binaural interaction (BI) in the auditory brainstem response (ABR) signifies the waveform discrepancy between a binaural response and the sum of monaural responses. A typical BI in the human ABR is the DN1 that shows binaural amplitude reduction occurring at wave V. By delivering sounds at one of three locations (either left, middle, or right) and also controlling attention modality, Ikeda (2015) found that the DN1 elicited by tone pips was vulnerable in comparison with the click-evoked DN1. This study examined the effects of sound delivery distributed randomly to the three locations on the DN1. The DN1 evoked by tone bursts was absent for both conditions of auditory and visual attention. The click-evoked DN1 was seen for any attention conditions while visual tasks reduced the DN1 amplitude. These outcomes support the contribution of high frequency sounds to the DN1 and also the facilitation of BI through auditory attention.

OR1834
Phenomenally transparent and non-transparent color spreading in flank transparency displays

Eiji Kimura *CHIBA UNIVERSITY, Japan*

Flank transparency refers to illusory color spreading that is induced when narrow colored flanks are added to segments of black lines located within a virtual region. Illusory percept has been characterized as a colored transparent filter over the lines and background. We investigated luminance and color conditions that induce color spreading. We used green or achromatic flanks and systematically changed luminances of the flanks, lines, and background. Observers rated subjective certainty of color spreading using a five point scale. Results showed that when the stimulus satisfies the luminance and color conditions for perceptual transparency, strong, phenomenally transparent, color spreading was observed. These results suggest that perceptual scission plays a central role in producing color spreading. Moreover, non-transparent type of color spreading was also found when the luminance condition was not appropriate for transparency. These findings suggest the contributions of different visual processes to color spreading in flank transparency displays.

OR1835
Review of experimental phenomenological studies about animacy perception as perceptual organization

Hidemi Komatsu (1), Atsushi Sakai (2), Naoe Masuda (3) *1. Keio University, Japan; 2. Meisei University, Japan; 3. Shizuoka Sangyo University, Japan*

Gibson (1979) asserted, "Psychology begins with the division between the inanimate and the animate." Animacy perception is essentially important for animals (including humans). This presentation will report two experimental phenomenological studies about animacy perception. The starting point of this approach is our daily experience. This approach regarded observation and description as important. Verbal description was especially weighty. In both studies, each of participants observed several kinds of moving toy. As result, merely because a certain object imitated the appearance of a real animal, it was not necessarily perceived as an animal. It was really important that irregular changes happened during its movement. And when the object appeared to move spontaneously, at the same time, the object has intention to the other object. The perception of relation among objects was inseparable to perceive animacy. Animacy perception could be seen as a kind of perceptual organization.

OR1836
The Boundary of Holistic Processing in Appraisal of Facial Attractiveness

Chang Hong Liu (1), Wenfeng Chen (2) *1. Bournemouth University, United Kingdom; 2. Chinese Academy of Sciences, China*

This study investigated whether the way external features of the face are masked affects perception of facial attractiveness. The occluding shape of the masks was defined either by the face's facial outline or by a uniform oval. Participants rated attractiveness of the masked faces. Results showed that faces masked by the oval shape were rated more attractive. Additional mask manipulation confirmed that the effect was mainly due to occlusion of a larger area of the external features rather than to the oval's regular and symmetrical features. The effect demonstrates an interaction between processing of the shape of the mask and the internal features of the face. The effect of this holistic integration on appraisal of facial attractiveness is striking because the oval shape of the mask is evidently not a part of the face, but the edge of an occluding object.

OR1837
Development and Validation of the Career Readiness Scale using the Item Response Theory

Maria Felicitas M Mamauag, Eric B Amaladas *HELP University, Malaysia*

The Career Readiness Scale measures the capability of an individual to make appropriate career choices, taking into account the complexity of family, social, economic, and organizational factors that influence an individual's career development (Sampson, et al., 2000). Using a four-option agreement scale, it was designed to form part of a test battery meant for career exploration among students aged 14–23. Items were generated from reflection papers of students taking a Career Development course in a private university in Malaysia. Using an IRT approach called Partial Credit Rasch Model, findings show that the items fit the model obtaining infit values of 0.5-1.5, indicating the absence of redundancy, and unidimensionality of the construct measured. Confirmatory factor analysis further tested the two-factor structure (capability and complexity) to have a good fit, $\chi 2=163.28/df=19$, RMSEA=.09, PGI=.93, GFI=.92. However, it is recommended to run the response category

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analysis to further determine the preciseness of the scale.

OR1838
Geometrical Structure of Gaze Perceptional Space

Masaki Mori (1), Toshio Watanabe (2) *1. Keio University, Graduate School of Media and Governance, Japan; 2. Keio University, Faculty of Environment and Information Studies, Japan*

The previous study has shown that gaze perceptional space was not enough to be explained by linear regression equation in term of depth distance. This study replicated the geometrical structure of the space based on gaze perception. Estimating the lateral angle between gaze perceptional and physical spaces by linear regression equation, the constant and the regression coefficient range from −2.14 to −1.02 and from 2 to 2.11. Furthermore, the constant and the regression coefficient were not related to the depth distance. Estimating the depth distance between gaze perceptional and physical spaces by linear regression equation, the constant and the regression coefficient range from −31.9 to 67.86 and from 0.36 to 0.63. Furthermore, the constant becomes larger when the physical angle becomes larger, although the regression coefficient was not related to the lateral angle. These results suggest that depth distance in gaze perceptional space has an anisotropic property.

OR1839
Vection perceived by the wind

Kayoko Murata (1), Hidemi Komatsu (2), Shigeru Ichihara (3), Naoe Masuda (4), Shigeru Ishihara (1) *1. Tokyo Metropolitan University, Japan; 2. Keio University, Japan; 3. MEDIA EYE, Japan; 4. Shizuoka Sangyo University, Japan*

"Vection", known as the train illusion, is a visually induced self-motion perception. Observers on a stationary train occasionally perceive a forward movement when another train on a neighboring track moves into the opposite direction. Such perceptual phenomena have been mainly investigated in relation to the visual system. (Fischer & Kornmüller, 1930). However, various sensory modalities other than vision, e.g., auditory, vestibular, somatosensory, and proprioceptive systems, also contribute to the self-motion (Siegler et al., 2000). We recently confirmed that the self-motion perception could be elicited by cutaneous stimulation accompanied by vestibular inputs (Murata et al., 2014). All indices of perceived self-motion (latency, duration, & magnitude estimation) were modulated as a function of the wind speed which was applied to the participants' face. The function of perceived self-motion by facial cutaneous sensation indicated a peek velocity at 3.0m/s. We discussed this feature by assuming a possible interaction between cutaneous and vestibular systems.

OR1840
Perceptual contrast between two short adjacent time intervals marked by clicks

Yoshitaka Nakajima (1,2,4), Mizuki Matsuda (2), Erika Tomimatsu (3), Emi Hasuo (3) *1. Department of Human Science, Kyushu University, Japan; 2. School of Design, Kyushu University, Japan; 3. Faculty of Design, Kyushu University, Japan; 4. Research Center for Applied Perceptual Science, Kyushu University, Japan*

Stimulus patterns of two adjacent time intervals, T1 and T2, marked by clicks were employed in our previous study [Nakajima et al., 2014, Frontiers in Human Neuroscience]. The presence of a shorter T1 caused T2 to be overestimated when T1 was 80–280 [ms] and T2 > T1+200 [ms]. This overestimation was considered a perceptual contrast to be investigated further. Either T1 or T2 served as a standard time interval to which a separate comparison time interval was adjusted to be subjectively equal. Both T1 and T2 were varied from 100 to 600 ms. Results showed that T2 was overestimated by 60–140 ms in the same range of conditions as in our previous study. T1 was also overestimated, but to a smaller extent, when T2 was up to 200 [ms] and T1 = T2+100 or T2+200 [ms]. No significant underestimation of T1 nor T2 appeared. Contrast took place only as overestimation.

OR1841
Cognitive style and gender differences in wayfinding

Danna Naurzalina (1), Elizabeth Davletkalieva (2), Bakit Kulbayeva (3), Bibigul Almurzayeva (4), Olga Aymaganbetova (5), Olessya Magsumova (6), Elmira Amanbayeva (7) *1. Turan University, Kazakhstan; 2. Aktobe Institute of Advanced Studies Orleu, Kazakhstan; 3. K. Zhubanov ARSU, Kazakhstan; 4. K. Zhubanov ARSU, Kazakhstan; 5. al-Farabi KazNU, Kazakhstan; 6. Military Institution of Air Defence Forces named after twice hero of Soviet Union hero Talgat Begeldinov, Kazakhstan; 7. KazNAU, Kazakhstan*

The purpose of a study is examining connection between FI (Field independent)/ FD (Field dependent) and wayfinding, N=120, Me(age)=29.7, 57 males and 63 females. The questionnaire "How do you navigate in Almaty", Pointing Task (PT), Kim's modification of text structure test "Wolf", BSRI are used. Key landmarks for most of examinees (85%) are large trade houses and shopping malls. Males use in their navigation reference points and cardinal directions. Males tend to make less mistakes in wayfinding than females, significant correlation between results of PT and sex is (.274**). The hypothesis about influence of cognitive style on wayfinding wasn't proved. Field independent (FI) participant spend less time on testing compare with field dependant (FD) participant. Drivers show better results than pedestrian (−.359**). Thus, FI (Field independent) and FD (field dependant) participants has no significant differences in wayfinding. Males tend to make less mistakes in wayfinding than females.

OR1842
What is inspiration? : the perception of Thai univestiy students after watching inspirational movies

Narisara Peungposop, Dusadee Yoelao, Wilailuck Langka, Saran Pimthong *Srinakharinwirot University, Thailand*

The objective was to describe the meaning of inspiration as experienced by the students after watching the movies. The research design was qualitative methods. Data was collected from 40 students selected from criteria. Structured interview guidelines with open-ended questions were used, which was then analyzed using grounded theory coding procedure. The results revealed the impact of watching inspirational movies on state inspiration. The participants described state inspiration as a force towards goal, force inside oneself, and force from love of others. The force towards goal was described as an infusion of some ideas that suggest some feeling to create moving forward to the goal. The force inside oneself was described as a feeling of understanding meaning of life, hope, and not admit defeat. The force from love of others was described as feeling energized from love of others.

OR1843
The Role of Racial Membership in Change Detection of Facial Expressions

Tan Serene, Zi Chien, Cameron Teoh *HELP University, Malaysia*

Human faces are visual stimuli of great social and biological significance, which can convey abundant source of social information (such as one's emotional state and ethnicity). However, little is known about the interaction effect of emotion and race on face perception. In a change detection paradigm, this 2 × 3 × 3 mixed-subject experimental study examined racial membership in change detection of emotional faces. 87 undergraduates of Chinese, Malay, and Indian were recruited, who were exposed to a series of flickering images of faces from the three races, and their change detection response time was measured. The findings revealed that Chinese faces were detected faster than Malay and Indian faces whereas changes from neutral to happy faces (especially of Malay) were detected faster than changes from neutral to angry faces, showing the reversal of expected results. One explanation is that happy expression (smiling) is prioritized for maintaining social relationships.

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OR1844
The Bogart Illusion Inverted

Tzuyao Tai, Akiyoshi Kitaoka *Ritsumeikan University, Japan*

Humans are accurate in judging others' gaze direction. However, some studies demonstrated gaze-shift illusions including the Bogart illusion (Sinha, 2000) where apparent gaze direction is reversed by reversing the contrast of portraits. We examined the effects of orientation and of presentation duration on this illusion. Results showed that the illusion was rarely perceived when faces were upright, but it was frequently seen with inverted faces. Moreover, this illusion was observed with short exposure duration, while it was rarely reported with long exposure duration. In the second experiment, instead of a two-alternative forced-choice task, the perceived duration of each gaze direction was measured. Results replicated that of the first experiment where face inversion increases the strength of Bogart's illusion, but in the second experiment the illusion was perceived consistently even in a longer exposure duration. These findings suggest that both luminance information and configural information may be involved in gaze processing.

OR1845
Collinear Gabor structure and visual selective attention

Chiahuei Tseng, Jiayu Leung *University of Hong Kong, Hong Kong*

Perceptual grouping is the structured wholes constructed from some foundation elements, and it was suggested to be an overlooked factor that modulated our selective attention (e.g. Jingling and Tseng, 2013). In this study, we tested whether the previous claim based on simple bars was generalizable to other types of stimuli (i.e. static Gabors, sinusoidal grating patches with gaussian intensity modulation). We generated visual search displays that consisted of 21 x 27 identical gabor elements except one randomly selected column (distractor) was replaced with gabors of orthogonal orientation. Observers looked for a slightly tilted target gabor and reported its orientation (discrimination task). Although the target location was independently selected of the distractor column, we discovered participants' search performance dependent on whether the target location on or not on the distractor. This additional evidence suggested that search impairment imposed by collinearity is a general observation across stimuli type.

OR1846
Assumptions about illumination account for differential interpretations of "the dress" stimulus

Pascal Wallisch *New York University, United States of America*

There has been considerable interest in a stimulus display producing starkly differing interpretations among observers, some of which perceive the image (of a dress) as white and gold, whereas others see it as black and blue, with only a few of them switching. Such a stimulus display is unprecedented in the domain of color vision. To date, no compelling explanations have been proposed. Here, we present data from a large global sample of 8000 participants that suggests that lifestyle choices (e.g. those due to circadian type) shape experience, which in turn determines illumination assumptions, which in turn can account for why individuals see "the dress" in one way or the other. We frame our explanation in terms of a Bayesian framework.

OR1847
Tap ahead of time: its relationship with tapping variability

Junkai Yang (1), Feiyi Ouyang (1), Linus Holm (2), Liang Zhou (1), Yingyu Huang (1), Lingyu Gan (1), Huizhen Chao (1), Mengye Wang (1), Sheng Zhang (4), Bo Yang (3), Junhao Pan (1), Xiang Wu (1) *1. Sun Yet-sen University, China; 2. Umea University, Sweden; 3. Anhui University of Traditional Chinese Medicine, China; 4. Anhui Medical College, China*

Researchers have long puzzled over the phenomenon that people tend to tap ahead of time: when synchronizing movements (e.g., finger taps) with an external sequence (e.g., a metronome), their taps slightly precede event onsets, representing the well-known negative mean asynchrony (NMA). Here, by presenting twenty-four metronome-tapping data, we found that the NMA decreased as tapping variability (the standard deviation of asynchrony) deceased, as long as synchronization could be performed (indicated by the negative lag-1 autocorrelation); regardless of the number of sequence events, the length of inter-onset intervals, the modality of stimulus presentation, stimulus duration, and presenting auditory feedbacks or not. This relationship between the NMA and tapping variability demonstrates an intrinsic link between these two primary measures in sensorimotor studies, and suggests that the NMA is associated with people's attempts to minimize their tapping variability. Keywords: Sensorimotor; synchronization; asynchrony; NMA; variability

OR1848
Attention enhances perceived contrast by attenuating the surround suppression

Liufang Zhou, Shena Lu, Yongchun Cai *Zhejiang University, China*

We studied how attention enhanced perceived contrast when the stimulus is embedded in a suppressive surrounding. Two vertical-oriented gratings were presented side-by-side on the screen, surrounded by a half-ring grating oriented either vertical (the reference side) or horizontal (the test side). Subjects were asked to judge whether the contrasts of the two target gratings were the same. Attention was manipulated by a dual-task paradigm: the contrast equality judgment task was poorly or fully attended according to whether a concurrent central letter-discrimination task was performed or not. We find that under fully attended conditions the perceived contrast of the reference stimulus is significantly enhanced compared with that under the poorly attended conditions, demonstrating a strong boost of the perceived contrast when attention is available. This may due to the attentional attenuation of suppression from the surround instead of direct signal enhancement.

POSTER PRESENTATION

P2396
Lightness perception in depth inversion

Tetsuya Arai (1,2), Yuka Igarashi (2), Keiko Omori (2), Yasunori Aizawa (2,3), Tomohiro Masuda (1) *1. Bunkyo University, Japan; 2. Kanagawa University, Japan; 3. Nihon University, Japan*

Many studies have examined perceived depth reversals such as the Mach book and reverse perspective, but few have reported other visual phenomena that simultaneously occur with depth inversion (e.g., size change or lightness change). We experimentally measured lightness change using the matching method when depth inversion was observed. We used a three-dimensional concave object that was composed of three paper boards and could be perceived as two depth arrays. Participants observed this object as concave or convex and judged the lightness of surfaces in each observation. Our results show that lightness of surfaces clearly changed as a function of depth perception. We discussed this relationship of depth perception and lightness perception in terms of perceptual organization.

P2397
Sounds and flashes affect the stability of visual time perception differently

Riku Asaoka, Jiro Gyoba *Graduate School of Arts & Letters, Tohoku University, Japan*

The present study examined inter- and intramodal modification effects on visual time perception. The tones and checkerboards were presented before and after the visual target stimulus as non-targets in Experiment 1 and 2, respectively. The participants were asked to reproduce the perceived duration of target visual stimulus using key pressing. The results showed the perceived duration was shorter with 0 ms inter-stimulus interval (ISI) between

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the target and non-targets than in the absence of non-targets in both experiments. The tones increased the stability of visual time perception in the 0 ms ISI condition compared to the target-only condition (Experiment 1). However, the checkerboards decreased stability in the 0 ms ISI condition in comparison with the target-only condition (Experiment 2). These results suggest that the inter- and intramodal modification effects on visual time perception are based on different mechanisms in the time reproduction task.

P2398
Oculomotor markers of stimulus importance

Yulia V Bessonova, Alexander A Oboznov, Nickolay A Almayev, Olga V Murasheva *Institute of Psychology, Russian Academy of Sciences, Russia*

Application of psychophysiological indicators could improve the reliability of measurement. Modern eye-tracking equipment has it's several advantages: it's a non-invasive, remote, precise and fast. The issue remains unresolved whether the attention is determined within features of the stimulus or importance to subject. The study aims to identify the markers of stimulus importance. Fundamental differences were found in gaze patterns determined by composition and subjective meaning of stimulus. The path of gaze is determined by composition for free viewing - first fixation usually occurs at the center, color of stimulus isn't essential compared to the size, location and grouping stimulus. Composition stops to affecting to eyemovement in case of viewing important stimulus, attention is fixed on it. Regardless of individual strategy of looking, the stable markers of stimulus importance have been found, namely as the increased amount of fixations, gaze returns, speed parameters of saccades as the decrease of sensory latency.

P2399
Bidwell effect under various chromatic illuminations and different eccentricities

Remigijus Bliumas, Algimantas Svegzda, Rytis Stanikunas, Ausra Daugirdiene, Henrikas Vaitkevicius, Vaiva Kulbokaite *Vilnius University, Lithuania*

The Bidwell effect is a convincing colour illusion phenomenon. The objective of the study was to investigate colour perception of Bidwell effect under various chromatic illuminations (red, amber, green, blue and neutral D65) and different eccentricities (0 deg. and 7.4 deg.). The disk rotated at 6 Hz frequency. Presentation time was 4 ms for colour stimulus, 81 ms for black and for white parts each. The chamber was illuminated by computer controlled quadrichromatic solid-state lamp, containing primary light emitting diodes. Asymmetrical colour matching procedure was used. The results were examined by means of perceived hue, value and chroma of image and afterimage. Such tendencies were observed: chromatic illuminant does have an influence on perceived hues of images and afterimages, chroma and value of image and afterimage are dependent on eccentricities, the same stimulus images and afterimages are roughly complimentary colours for all of them.

P2400
Changes in visual sensitivity for luminance and chromatic stimuli during eye movements

Doris I Braun (1), Karl R Gegenfurtner (1), Alexander C Schütz (2) *1. Giessen University, Germany; 2. Marburg University, Germany*

Eye movements change visual sensitivity continuously. Saccadic suppression, an active process following a specific time course around saccades, was reported to impair selectively the perception of luminance stimuli at low spatial frequencies, while during pursuit the visibility of isoluminant colored stimuli was improved. We compared sensitivity to flashed horizontal luminance and isoluminant colored lines before and during saccadic and pursuit eye movements. All of our observers showed a marked saccadic suppression for both luminance and for isoluminant targets, for saccades to both stationary and moving targets. The magnitude of the suppression for color was about half of the one for luminance. During pursuit initiation only color sensitivity increased. Results were similar in darkness and under daylight conditions. We conclude that saccadic suppression affects vision in a more general manner than previously thought. Whatever sensitivity improvement there is during pursuit eye movements, is suspended during the initial saccade to moving targets.

P2401
Can Taiwanese political parties be categorized by face? Exploring the effect of perceived age on categorization accuracy

Chien-Kai Chang, Sarina Hui-Lin Chien *Graduate Institute of Neural and Cognitive Sciences, China Medical University, Taiwan*

Rule and Ambady (2010) found that US college students can differentiate Republicans and Democrats by face. The present study aimed to explore whether the two Taiwanese political parties, Kuomingtang (KMT) and Democratic Progressive Party (DPP) can also be differentiated by face. An age rating task was first conducted to control the effect of perceived age on categorization accuracy. 24 participants rated the perceived age of 100 gray-scale photos of candidates from the KMT and DPP. In the main categorization task, 33 participants categorized 80 photos as either KMT or DPP. The photos were presented in four age blocks (under 40, 41–45, 46–50, over 50). Results showed that across all blocks, participants could categorize KMT and DPP with accuracies significantly higher than chance. Moreover, for KMT candidates, the perceived age showed a mild positive correlation with categorization accuracy, while the perceived age of DPP showed a mild negative correlation instead.

P2402
Integration or separation in the processing of facial properties

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Human faces contain information about identity, facial expression, sex and race. Whether separate neural units in the brain deal with these face properties individually or whether a single neural unit processes in parallel all aspects of faces is currently debated. In a theoretical framework we tested whether a system can deal with identity in combination with sex, race or facial expression using the same mechanism. With dimension reduction we describe how the representational facespace organizes face properties when trained on different facial aspects. When trained to learn identities, the system not only successfully recognized identities, but also was immediately able to classify sex and race, suggesting that no additional system for the processing of invariant properties is needed. However, training on identity was insufficient for the recognition of facial expressions. We provide a theoretical approach on the interconnection of invariant properties and the separation of variant and invariant properties.

P2403
Crucial period of online vision for grasping control is modulated according to movement duration

Takao Fukui (1), Toshio Inui (2) *1. Research Institute of National Rehabilitation Center for Persons with Disabilities, Japan; 2. Otemon Gakuin University, Japan*

The current study examined how movement duration modulates use of early phase online vision for reach-to-grasp movements by manipulating the available viewing time and setting two movement times (500 ms vs. 1000 ms). Liquid-crystal shutter goggles were used to manipulate the duration of available online vision during the movement. We found that non-availability of early phase online vision

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(after movement initiation) affected the grip configuration (i.e., inducing a larger peak grip aperture) even in the shorter movement duration and that the increase in the peak grip aperture was depending on the non-available viewing duration. The influential period for grasping control shifts to an earlier time when movement time is shorter (i.e., from approximately 214 to 106 ms after movement onset), indicating a flexible visuomotor mechanism according to the movement duration and the available online vision.

P2404
Sound symbolism modulates perceptual judgment on dynamic events

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Phonetic information can be associated with particular meaning in mental processing (e.g., bouba/kiki effect). This association is called sound symbolism. Onomatopoeia that mimics sensory information by its sound (e.g., bow-wow) is a kind of words with sound symbolism. Onomatopoeia has abundant information based on perceiver's own sense and experience of motion situations as well. However, it is unclear whether onomatopoeia can affect perceptual judgment on dynamic events. In this study, we investigated this issue by using Japanese onomatopoeia and several dynamic phenomena such as stream/bounce perception and visual representational momentum (visual RM). In the experiments of stream/bounce perception, onomatopoeia representing "strong crash" enhanced the proportion of judging "bounce" and onomatopoeia representing "go through" increased the proportion of judging "stream". In the experiments of visual RM, onomatopoeia representing "stopping" weakened the magnitude of visual representational momentum. These results suggest that sound symbolism of onomatopoeia can modulate judgment on dynamic event perception.

P2405
fMRI study of the visual search for self-controlling moving objects

Kazume Horita, Takako Yoshida *Tokyo Institute of Technology, Japan*

It is reported that self-controlled moving object was visually searched faster than other-controlled one. To examine the relationship between attention, multimodal congruency, and sense of agency that "I," not "other," is controlling the object, we measured the BOLD signals for the self- and other- controlling objects. Subjects searched for a target among other objects moving around randomly, while freely controlled the target position using trackball. Self-generated real-time control (SE) and the target position following the other individual's trackball instead of the participant's trackball (OT) were two of four conditions. The comparison between OT and SE showed activation in the bilateral inferior parietal cortex and corpus callosum. These results suggest that the bilateral inferior parietal cortex is involved in the monitoring of the multimodal congruency of the predicted and actual outcomes and recruited when they don't match. The relationship between this monitoring function and visual attention is discussed in this report.

P2406
Adults with Asperger syndrome or high functioning autism are less proficient in identifying point-light biological motion displays

En-Yun Hsiung (1), Shu-Fei Yang (1), Sarina Hui-Lin Chien (1), Yu-Hsiu Chu (2), Cheng-Xin Ye (2) *1. Graduate Institute of Neural and Cognitive Sciences, China Medical University, Taiwan; 2. Department of Physical Therapy, China Medical University, Taiwan*

Recent studies demonstrated impaired biological motion perception in adults with autism spectrum disorder (ASD). Using point-light displays portraying human actions interacting with (i.e., paddling) or without objects (i.e., running), the present study examined the action identification performance in healthy adults and adults with Asperger (AS) or high functioning autism (HFA). Twenty AS/HFA and 30 healthy adults (aged 19–43 years) viewed 20 different biological motion clips and their naming responses and speed were recorded. Results revealed that AS/HFA adults had a lower number of correctly named items (M=14.95, SD=3) than that of healthy adults (M=15.67, SD=1.54), and longer response time (M= 4948ms, SD= 1449) than that of healthy adults (M=3615ms, SD= 1130). Moreover, both groups were better at naming the actions without objects (AS/HFA: 8.7, healthy adults: 9.5) than those with objects (AS/HFA: 6.3, healthy adults: 7.4). In conclusion, adults with AS/HFA were less proficient in identifying point-light biological motion actions.

P2407
The effects of the three attributes of color on visual roughness perception

Takashi Inaba *Nihon University Graduate School of Social and Cultural Studies, Japan*

The aim of this study is to clarify the relationship between the attributes of colors and the perception of textural roughness. Participants evaluated the linguistic images of the degree of roughness evoked from color stimulation and the selections of corresponding roughness by actually touching the roughness samples. As a result, a "rough" evaluation and selection of the deeply wrinkled sample were made for the low-chroma and the medium-gray colors, whereas a "smooth" evaluation and selection of the superficially wrinkled sample were made for the high-chroma and the black-and-white colors. Furthermore, a positive correlation was recognized between these two evaluations.

P2408
Upon the validity and reliability of Munsell color system: Revisit

Mitsuyo Ito *Tokushima University, Japan*

The color atlas by Munsell system(HV/C) is assmbled for each surface color to be locating at equal distance in uniform color space. The color system of CIE(a*b*), derived from Adams' stage theory of color vision, is also constructed as uniform color space by three primary colors mechanism, two oppnent color proceses, Stevens' power law function and the adaptation to illumination. If we appply 10 basic hues of Munsell system onto CIE(a*b*), we expect that the perceptual distance between those colors appears to be equal, but it is not always the case. The situation then raises a question, whether the color space of Munsell system is perceptual uniform or not. Considering that Adams' theory of color vision has kept most sophiscated at present, we must revisit for Munsell system to be assume the perceptual uniformity of color space.

P2409
The effect of visual movement information on auditory sensitivity

Yoonbum Kang, Shinya Saida *Kanagawa University, Japan*

Audio-visual interactions between visual movement intensity and the sensitivity of the auditory modality have been little examined. We investigated the effect of visual movement information on auditory sensitivity by using videos of clapping hands. We manipulated the visual movement intensity of the clapping hands (low, high, only fixation point) and the stimulus onset asynchrony (SOA) between visual and auditory stimuli. We asked the participants to indicate whether a clapping sound was presented when they observed a video of clapping hands together with the sound of noise. The volume of the clapping sound was adjusted to each participant's threshold value. The results indicated that the longer the SOA between visual and auditory stimuli, the lower the detection rate of the clapping sound; and this rate was even lower when only the fixation point was presented instead of the clapping hands. Therefore, this suggests that the visual movement information affects auditory sensitivity.

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P2410
Visual attention affects the fusional limit of a peripherally located object when using a Posner spatial cueing task

Ken Kihara (1), Hiroshi Ono (2) *1. Kagoshima University, Japan; 2. York University, Canada*

We rarely experience diplopia in daily life although most objects have non-zero retinal disparity. We investigated whether visual attention affects Panum's fusional area where we can perceive an object with retinal disparity as single. For this purpose, we used a Posner spatial cueing task where attention is automatically oriented to a peripheral location by salient exogenous cues. The task was to rapidly identify the location of a vertical-line target with disparity and report whether the line was single or double. The results from seven subjects showed that the size of Panums' fusional area was larger when the target was presented at the uncued rather than at the cued location. This tendency could be seen independent of the distance between the cued and the uncued locations. These results suggest that a peripherally located object with disparity is less likely to be perceived as diplopic when located outside the focus of attention.

P2411
Effects of lip color on perceived lightness of human facial skin.

Yuki Kobayashi, Soyogu Matsushita, Kazunori Morikawa *Osaka University, Japan*

Adjacent colors can have two conflicting effects: color contrast and color assimilation. In facial makeup, these effects may affect perceived lightness of facial skin. We investigated whether lightness and color of lips have those effects. The standard stimuli were a female face with four versions of simulated lip colors (natural, lighter, darker and more reddish). The comparison stimuli were the facial area between the upper lip and the eyes at nine levels of lightness. Perceived lightness of the skin of each standard stimulus was measured by the method of constant stimuli. The results showed that the darker lips darken the perceived skin color whereas the more reddish lips lighten it, suggesting that both color contrast and color assimilation can occur in the human face. These findings indicate the influence of lipsticks on perceived skin lightness.

P2412
Role of the locus coeruleus-noradrenaline system in attention direction for visual change detection: a pupillometric study

Hijiri Kodama, Ken Kihara, Sakuichi Ohtsuka *Kagoshima University, Japan*

Humans sometimes fail to notice obvious changes in objects in the field of view when the changes are unexpected. Failure to discern the change occurs because the object was not being attended to. This phenomenon is known as inattentional blindness. It is suggested that the activity of the locus coeruleus-noradrenaline (LC-NA) system is involved in regulating attention. This study uses measured pupil response and the inattentional blindness paradigm to investigate the temporal dynamics of the LC-NA system before and after the visual detection of change. Results show that pupil dilation was observed four seconds before the detection of change on average. This finding suggests that the LC-NA system plays an important role in orienting attention to visual changes in objects and awareness of the changes.

P2413
Effect of Body Posture on the Interpretation of Cast Shadows

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We investigated the light-from-above (or light-from-left) assumption in interpreting cast shadows, especially, which of the head or gravitational coordinate system the assumption depended on. Our stimuli included five grey disks and four blurred shadows between the disks arranged vertically or horizontally. With four observation postures (sitting, lying-on-left, right and back), we measured relative strength of perceptual correspondence between the center disk and the adjacent cast shadows. The results showed a light-from-above bias in the head coordinate system under the sitting, lying-on-left and lying-on-back conditions and a light-from-left bias only under the sitting condition. We conclude that the light-from-above (or light-from-left) assumption is effective when the head and gravitational coordinates are consistent. When the head is not upright, the assumption with the head coordinate system becomes less effective possibly because the visual system refers to both coordinate systems.

P2414
Archetypes of 3D Shape Design: An Analysis of Bottle Shapes Using Spherical Harmonics

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This study examines the relationship between 3D product designs and the impressions of the shape used to extract prototypical 3D shapes of package designs that evoke specific impressions. We refer to such prototypical shapes as "archetypes." To quantitatively analyze a 3D shape design, we used a combination of 3D scanning and spherical harmonic expansion (SPHARM). The SPHARM is a parametric surface description that uses spherical harmonics as basis functions. We scanned the 3D surface design data of 22 body soap bottle shapes with an optical 3D digitizer. By using SPHARM, each of the surface data was represented as a multivariate vector. Participants rated their impressions of the 3D printed bottles on a seven-point semantic differential scale. The bottle shape corresponding to each of the impressions within the impressions set was reconstructed using the inverse SPHARM transformations. Consequently, the morphological features of the reconstructed bottle shapes should reflect our 3D archetype.

P2415
Effects of repetition on the isochronous perception of sounds at different lengths

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We investigated how the isochronous perception of successive sounds would be modulated by the length of these sounds. Three sounds, each one being short (S) or long (L), were successively presented in Experiment 1. The middle sound's onset had to be presented earlier in the SLS than the LSL sequence so that the three sounds were perceived as presented isochronously, indicating delayed perception of the onset with the lengthening of the sound. A short sound and a long sound were alternated repeatedly in Experiment 2. This repeated sequence was perceived as isochronous when all sounds were aligned at physically equal inter-onset intervals. Would the same onset delay as in Experiment 1 occur, the repeated sequence should have been perceived as isochronous when the SL interval was physically shortened relative to the LS. The effects of sound length seemed to be canceled out when the presentation of intervals was repeated.

P2416
Effect of continuity in visual search.

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Salient stimuli usually capture attention. However, our previous studies showed a masking effect by a salient collinear distractor on a local target. The salient collinear distractor was formed by grouping laws similarity and continuity. This study aimed to test whether continuity is enough to elicit the masking effect. Experiment 1 randomized the orientation of

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background bars. Experiment 2 made the distractor curved. Both experiments reduced similarity in background (Experiment 1) or distractor (Experiment 2). Participants discriminated the orientation of a gap on or off the distractor. Results showed that in both experiments a target on the distractor was discriminated faster and more accurately than that in the background. Our results suggest that good continuity facilitates visual search. Thus the masking effect by a salient collinear distractor reported in our previous study must be caused by the grouping law of similarity.

P2417
The role of blur in the motion illusion induced by the luminance gradient in stationary images

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Repeating patches of luminance gradient produce illusory motion in a stationary image. This study examined whether temporal changes in the amount of blur of the image are associated with motion illusion. In Experiment 1, participants observed illusory figures through a zoom lens and operated the focus ring. The results showed that the perceived location of the patches shifted toward a certain direction as the figure was defocused, and returned as it was focused again. In Experiment 2, participants observed the figures that were blurred with digital photo editing techniques. The results demonstrated that the perceived location of the patch shifted as the blurriness increased. We speculate that the accommodation of the eye and/or the change of spatial frequency that the visual system extracts are analogously associated with this kind of illusion.

P2418
The Composition by Line Drawing has a Facilitatory Influence on Fick Illusion?

Kazuhiko Mori *Akita University, Japan*

The Fick illusion(vertical-horizontal illusion) is the tendency for observers to overestimate the length of a perpendicular line relative to a horizontal line that has the same length. The present study examined the facilitatory or inhibitory influence on Fick illusion in visual-kinetic judgments by perpendicular line-drawing relative to the standard lines provided on frontal parallel plane. The participants were 80 undergraduates. The within-subjects design experiments were carried out which contained the visual-kinetic judgment conditions by ascending perpendicular line-drawing and the normal visual judgment conditions by method of limits with ascending series. Preceding judgment condition divides the participants in halves. Standard line lengths are 3 conditions, and T direction conditions have 12 angles. The results show that the composition by line drawing has a facilitatory influence. We discussed the representational momentum in the visual-kinetic judgment by reference to the preceding reports on the method of adjustment and the haptic Fick illusion.

P2419
The factors determining the appearance of Navon figures: the effects of exposure frequency and size range of stimulus.

Takashi Murakami *Ritsumeikan University, Japan*

We examined the effects of stimulus size range (low, medium, and high) and exposed frequency (early and late phases of trials) on the appearance of Navon (1977) figures, which usually alter between the global and local perceptual appearances. In the experiment, 18 participants were asked about which local or global feature of each stimulus they perceived more clearly. We determined the stimulus size at which global responses were replaced by local ones, i.e., the local threshold. The results showed that the local thresholds obtained were about 19°, 15°, and 12° for the large, medium, and small stimulus ranges, respectively, and the local threshold in the late phase of trials was lower than that in their early phase. We concluded that the Navon figure was organized by stimulus size itself and by stimulus size range and was changed into local appearance for prolonged exposure to the figure.

P2420
The effects of location shift in object-centered and ego-centered coordinates on the boundary extension in visual and motor task

Takuma Murakoshi, Eiji Kimura, Makoto Ichikawa *Chiba University, Japan*

We investigated the effects of location shift in two different coordinates on the boundary extension using visual and motor tasks. On each trial, after a partial image of a natural scene was presented (observation phase), participants pointed the boundaries of the remembered area on the full-size image (test phase). Before the presentation in the test phases, the center of the image was shifted according to different coordinate systems. In the first condition, the image center was shifted leftward or rightward in ego-centered coordinates. In the second condition, the image center was shifted leftward or rightward in object-centered coordinates. Results showed that the remembered area was narrower in the second than in the first condition and shifted toward the image center of the full size image. These findings suggest that the boundary of the remembered image is determined according to object-centered coordinates in both the visual and motor systems.

P2421
The Facing Bias in Point-Light-Walkers which Vary in a Walking Speed

Ko Nakamura *Hokusei Gakuen University Junior College, Japan*

This study investigates the facing bias in point-light-walkers which were made from video clips recorded from right behind the walkers walking on a treadmill at speeds of 2, 3, or 4 km/h. Those original point-light-walkers were arranged by reversing the motion of the upper, lower, or entire body. 115 observers were asked to judge whether each point-light-walker was facing them or facing away from them. Although the facing bias was not so strong as reported in previous studies, a slight facing bias was shown, especially for the original point-light-walkers at the speed of 4 km/h (31% of judgments), comparing to another group of original walkers at the speed of 2 or 3 km/h. Results also showed that the judgments by observers were largely dependent on the motion of lower parts of the body.

P2422
Temporal-order judgment for the unconscious visual target

Shun Nakano, Ishihara Masami *Tokyo Metropoliatan University, Japan*

Recent findings indicate that participants are capable of reacting to a given visual target without consciously experiencing the target. The present study aimed to elucidate the temporal characteristics of detecting such an unconscious visual target. Using a visual backward masking paradigm, we measured participants' responses for the masked target. Four circular masking stimuli were simultaneously presented twice in a trial sequence and a circular target stimulus was presented just before the presentation of either masking stimulus. Participants were required to indicate the temporal order (1st or 2nd) of the target presentation (timing-judgment task) and to indicate the spatial location of the target (location-judgment task). The results showed that correct responses of the 2nd target were significantly higher than those of the 1st target in the both tasks. RTs did not significantly differ from each other. The importance of being explicit to the unconscious visual target for selective responses was discussed.

P2423
Auditory scene context facilitates visual object recognition

Ryosuke Niimi, Takahiro Saiki, Kazuhiko Yokosawa *University of Tokyo, Japan*

We examined whether sound consistent with visual scene (e.g., ocean wave sound for a beach scene) acts as context and facilitates recognition of object in the scene. In each trial, participants heard 4 sec sound and observed

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a scene picture briefly presented at the end of the sound. The task was to name the central object in the picture. Naming accuracy was higher for trials with scene-consistent sound than for trials with scene-inconsistent sound, namely, the context provided by the sound stimuli facilitated visual object recognition. However, this context effect emerged only when all the visual scenes were comprised of objects and consistent background. In other two experiments in which inconsistent visual scenes (e.g., a deer on theater background) were used, the auditory context effect was absent. These results suggest that auditory scene context does not work independently but constitutes modality-independent scene context in tandem with visual stimuli.

P2424
Visual attention affects the fusional limit of a centrally located object when using an attentional blink task

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It is possible that visual attention influences the range of disparities within which fusion occurs (Panum's fusional area). To investigate this, we used an attentional blink task where two targets (T1 and T2) are embedded among distractors, sequentially presented. In this task, attention to T2 is weak when stimulus onset asynchrony (SOA) between the two targets is 200–500 ms. T1 was an arrow and T2 was a vertical line with disparity, presented at an SOA of 100, 300 or 700 ms. Participants were asked to identify the direction of the arrow and to report whether the line was single or double. The results revealed that diplopia occurred more frequently when SOA was 300 ms, suggesting that paying attention to a centrally located object contributes to the fusion of binocularly seen stimuli.

P2425
Psychophysical Experiments on Hardness Sensation of an Object from the Sounds of Tapping Created by a White Cane

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Previous researches suggest considering the roles of both auditory and tactile information from the white cane in determining objects' hardness is necessary. In this study, we examined the ability of people to perceive the hardness of an object solely through the tapping sounds of a white cane using a method of magnitude estimation. We used three types of handgrips to create different sounds of tapping on an object with a cane. And there are two types of sound conditions: 1) playback of recorded tapping sounds and 2) sounds produced on-site by tapping. The participants were sighted university students wearing eye masks and totally blind who walk independently with a white cane. The results showed that both participants were able to accurately judge the hardness of an object by using auditory information. And it was suggested that sound information was more dominant than tactile information in cross-modal interactions on hardness sensation.

P2426
The effect of intentionally and unintentionally exposure to an odor

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In this study, whether repeated presentation of an odor induced the modification of the odor liking when participants were exposed to the odor intentionally or unintentionally was examined. In exposure phase, the color matching task was done by using three kinds of shampoo with different odors, and color samples. Each of shampoos was presented 3, 6 or 9 times respectively. At this task, intentional exposure group was asked to pay attention to the odor of shampoo, as well as its color, while unintentional exposure group was not required to pay attention to the specific odor. As results, unintentional exposure group showed higher preference to exposed odors than intentional exposure group, however, in both groups exposed odors were not more liked than unexposed odors. The intentional repeated exposure might lead a habituation to the odor, and then the spoiled interesting in the odor may affect odor liking.

P2427
Dependence of effects of odors on the perceivers' preference: arousal and relaxing effects

Satoko Ohtsuka *Saitama Institute of Technology, Japan*

In spite of previous psychological and psychophysiological efforts, it is still controversial how a specific odor affects mood. In the present study we examined the odors' arousal and relaxing effects with basing on perceivers' preference. Olfactory stimuli employed were Rose absolute, Basil, Bergamot, and Cypress oils. Twenty undergraduate students smelled each stimulus; before and after smelling they responded to a self-report mood scale. Besides the participants rated how much they liked each odor. As a result any stimulus did not produce consistent effect. Analysis based on participants' preference, however, showed systematic patterns of effect. That is, liked odors generally raised the arousal ratings while remained the relaxation ratings unvaried. In contrast, unliked odors remained the arousal ratings unvaried while reduced the relaxation ratings. These strongly suggest that the odor's effect on mood depends on the perceivers' emotional response to each.

P2428
Range of perceptual simultaneity and frequency separation between two pure tones

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We previously reported that the range of perceptual simultaneity for two pure tones can be expressed by the two segmented regression lines as a function of frequency separation of the tones: decreasing in close frequency separation, and increasing in distant frequency separation. This study aims to examine the generality of this function. The stimuli used in this experiment are similar to those used in the previous study, except that the fundamental frequency was fixed at the level higher than that in the previous study. The range of perceptual simultaneity was measured by the use of the simultaneity judgment task at various frequency separations. Results show that the range of perceptual simultaneity can be fitted to two segmented linear regressions against the frequency separation, as in the previous study. These results suggest that the relation between the range of perceptual simultaneity and frequency separation was consistent regardless of the fundamental frequency.

P2429
Mismatch negativity (MMN) and the scale illusion: an electroencephalography (EEG) study

Kentaro Ono, Christian F Altmann *Kyoto University, Japan*

Mismatch negativity (MMN) can be elicited by any discriminable change of features in a tone sequence. The MMN is believed to occur as a result of comparison between the memory trace of previous tones and an incoming tone. In this study, we investigated whether the memory trace is created by physical properties or perceived representation of a tone sequence, using a tone sequence that produces illusionary perception of ascending/descending scale, known as "scale illusion". The order of presentation to the left and right ears was infrequently changed, used as a deviant. Dichotically presented ascending/descending scale, as a control, showed MMN in response to the deviant, while the illusion sequence did not show MMN. These results suggest that the occurrence of MMN is based on the representation of tones based on the perception, but not on the physical properties of a tone sequence.

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P2430
Perceptions of Negative Life Events among Japanese in Japan, Japanese in the U.S., and Non-Hispanic Whites in the U.S. Implications for Research and Practice

Mika Ozawa, Richard Mendoza *Alliant International University, United States of America*

Individual perceptions of negative life events tend to impact physical and mental health (Lazarus & Folkman, 1984; Marum, et al., 2014). A systematic examination of the role of perception and distress in negative life events was explored in a nonrandom sample of Japanese living in Japan, Japanese living in the United States, and Non-Hispanic Whites in the U.S. Demographic and life event questionnaires administered revealed, Japanese living in Japan have higher self-reported levels of distress than Japanese in the U.S., and Non-Hispanic Whites in the U.S., in response to environmental negative life events. Gender differences mediated the perception of distressing events. Given a recent rise in stressful world and economic events, this pilot study has important implications for the study and practice of psychology and the emergence of a new branch of study concerning the psychology of negative life events.

P2431
Quadri-stable perception induced by visual motion stimulus

Gerard B Remijn (1), Tatsuya Yoshizawa (2), Hiroaki Yano (1) *1. Kyushu University, Japan; 2. Kanazawa Institute of Technology, Japan*

Visual motion perception was investigated for a stimulus consisting of two identical bars that moved towards each other, coincided, and then moved away from each in continuing circular motion - with acceleration towards and deceleration away from the bars' point of coincidence. The stimulus thus combined streaming-bouncing motion with rotation based on ambiguous depth cues. Experiments showed that the stimulus induced quadri-stable perception, with percepts resembling motion patterns in a Polka dance: with steady fixation, the bars were either seen as moving through each other (streaming), as bouncing off of each other at the bars' point of coincidence (bouncing), or as moving in unison in circular motion, pivoting around an imaginary axis with spontaneous reversals in clockwise or counterclockwise direction. We investigated effects of the size of the bars, their separation and the speed of their motion on the appearance rate of the four percepts in this "Polka-dance" stimulus.

P2432
Bistable visual event perceptions can be altered by specific hand postures

Godai Saito, Jiro Gyoba *Tohoku University, Japan*

Bouncing perceptions can be induced more frequently when participants put their palms together in a stream/bounce display; however, it is unclear whether other hand postures affect visual event perceptions. To clarify this, we added conditions in which the hand positions were similar to that of palms together, such as a posture in which a rectangular brick was sandwiched between participants' palms and one simulating this, in which there was empty space left between the palms without the brick. Our results showed that only the condition in which palms were together led to a significantly enhanced bouncing perception, suggesting that proprioceptive, tactile, and haptic information regarding specific postures are important for altering visual event perceptions.

P2433
Warmth and Coolness of Colors in Color-Vision Deficient Observers

Keiko Sato *Kagawa University, Japan*

Some reddish colors appear de-saturated and brownish for color-vision deficient (CVD) observers with red-green color deficiency. In this study, warmth and coolness of colors and its color naming were compared between normal trichromat (NT) observers and CVD deuteranomalous and protanomalous observers, and the difference between NT and CVD observers in terms of cone-contrast in human color vision was analyzed. In this experiment, 32 color stimuli were sampled from the BCP (Berkeley Color Project), and seven CVD and 12 NT subjects performed two tasks. In the color naming task, they selected one color category from the 11 basic color terms. In the evaluation task, the same subjects rated warmth and coolness of colors. Overall, findings in the present study could help us better understand color naming and affective color meanings in CVD observers based on the cardinal channels of color vision.

P2434
Effects of changing-size and changing-disparity cues on visually induced self-motion

Yasuhiro Seya, Hiroyuki Shinoda *Ritsumeikan University, Japan*

Depth information is known to affect visually induced self-motion perception (vection). The present study examined the effects of changing-size and changing-disparity cues, which serve as cues of depth perception, on forward or backward vection. In three experiments, participants viewed optical flows with or without the two types of cues. In Experiment 1, participants pressed a button whenever they felt vection and, after each trial, rated the vection magnitude. In Experiments 2 and 3, the participants rated the perceived velocity and motion-in-depth impression, respectively. We found enhancements in vection and motion-in-depth impression when either or both types of cues were present, as compared with the vection induced by flows without them (Experiments 1 and 3). The perceived velocity did not change with cue condition (Experiment 2). These results suggest that the enhancement of vection caused by depth cues is related to impression of motion-in-depth rather than perceived velocity.

P2435
Length illusion on Benham's top

Kaori Shoji (1), Chiemi Miyata (1), Ken Kihara (1), Sakuichi Ohtsuka (1), Hiroshi Ono (2) *1. Kagoshima University, Japan; 2. York University, Canada*

A spinning Benham's top is known to induce apparent colors. When rotation speed is reduced to suppress most blur, we find that the perceived length of line segments at white to black transitions is clearly enlarged in addition to apparent color. We examine the length illusion in two ways; line segment length (using static line segments) and endpoint alignment (using other rotating marks outside the disk). The stimulus had three sections of black and white instead of one as is normal and rotated slowly, at 40 rpm, so apparent colors were faded. Results show that the line segment length was clearly illusory but the endpoint alignment was mostly correct, suggesting that the illusion occurs in a relatively higher perceptual stage; length or shape not location.

P2436
Temporal recalibration after motor-auditory delay is accompanied with reduction of auditory latency

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Perception of synchrony between one's own action and sensory feedback thereof can be recalibrated after exposure to a delay (temporal recalibration: TR). It is still in debate whether the mechanism of motor-sensory TR is universal across sensory modalities. We compared motor-visual and motor-auditory TR as to whether perceptual latency was changed along with the TR. Participants were exposed to a delayed visual or auditory feedback after voluntary action, and then they did a synchronous tapping (ST) task and a simple reaction time (SRT) task to the same stimulus during the exposure. The TR can be measured by the ST task and a change of perceptual latency can be measured by the SRT task. The

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results showed that the SRT became significantly shorter after exposure to motor-auditory but not to motor-visual delay, suggesting that the mechanism of TR is different between motor-visual and motor-auditory domains.

P2437
Gait perception of Bunraku puppets in the Japanese traditional performing arts

Shigemasa Sumi *Keio University, Japan*

In Bunraku Theater, three puppeteers manipulate a single puppet so in complete harmony that viewers are incapable of distinguishing the doll from human being. How Bunraku puppets and human actors are similar in processes of performance is examined. In normal walking, actors move heel and shoulder in coordination with each other and the scatterplots by the values of movements reveal shape of a loop (Sumi, 2000, Swiss Journal of Psychology. 59,126-132). Using the video images of human and puppets' gaits, I measured the values of their heel and shoulder movements in pixel size. The scatterplots from the results formed a loop, indicating similar trends in the regression-curves between human and puppets' gaits. The contribution rates of principal component analysis seemed to provide useful information about the personal traits of a puppet. We confirm that Bunraku puppet is really the performer possessing human attributes, moving in a world of its own.

P2438
Can humans be detected more accurately than other objects? Evidence from an ultra-rapid categorization task

Ryo Tachibana (1,2), Keiyu Niikuni (1), Toshiaki Muramoto (1) *1. Tohoku University, Japan; 2. Japan Society for the Promotion of Science, Japan*

Previous studies have revealed that object recognition is always affected by contextual scenes in the early stages of visual processing. For objects and context that correspond semantically (e.g., a ship and the sea), we can judge the situation in as little as 20 ms. The purpose of this study was to investigate how object/context interactions differ across object categories in natural scenes. Using an ultra-rapid categorization paradigm, we presented images of natural scenes comprising combinations of target objects (human, animal, and vehicle) and scenes (manmade and natural) to participants for a brief duration (13–93 ms). Results showed that the accuracy of detecting humans in manmade scenes was the lowest in the 13 m condition, and there were no differences in response time. These results indicate that humans were not detected accurately in this specific scene type, especially when the presentation length was very short.

P2439
Perceived duration can be shortened by scrambled structure of natural images, but is not affected by their impressions

Yuri Takahashi, Shu Imaizumi, Momoko Yamada, Haruo Hibino, Shinichi Koyama *Chiba University, Japan*

Scene structure can modulate time perception (Varakin et al., 2013). We investigated the effects of scene structure and impressions (appreciation, speed) on duration perception. Participants viewed upright, inverted, or scrambled natural images, and judged their duration (400 or 1600 ms). The temporal bisection point for each image type was computed. Results indicated that perceived duration for upright images was overestimated than for scrambled images. However, this effect disappeared when the background color was switched from white to black. This suggested that scene structure causes a perception of expanded duration; this effect can be nullified by other spatial properties such as luminance. Impressions however showed no effect. We examined perception for a relatively short time period (approximately 1000 ms); however, further research should investigate perception for a longer time period since perception of short durations differ from that of long durations (Lewis & Miall, 2003).

P2440
Apparent speed of object motion affects time perception

Airi Takase, Tadayuki Tayama *Hokkaido University, Japan*

It has been known that the duration of motion stimuli is perceived longer than that of static stimuli. The speed of object motion is considered as an important factor on time perception. Gorea & Kim (2015) revealed that the perceived duration of apparent motion increases with the apparent speed. If the perceived duration changes with the apparent speed, it will also change with the apparent speed of moving object in the induced motion. This study used two circles as stimuli. They were presented in higher and lower positions. Both of them were stationary, moved, or one of them moved rightward or leftward. Observers gazed at one of them and estimated the duration. The results indicated that not only the apparent speed of the attended object but also the existence of the non-attended object and its movement affected time perception.

P2441
Effects of negative emotional valence intensity on visual target detection performance

Yasuhiro Takeshima *Bunkyo Gakuin University, Japan*

Negative stimuli (e.g., angry face) are considered to be processed with higher priority. This priority is related to the rapid shifts of attention. However, the relationship between the processing superiority and negative emotional valence intensity has not been clarified. In this study, I investigated the effects of this intensity on visual target detection performance. In the experiment, participants were asked to discriminate one facial direction (first target: T1) and to detect one face (second target: T2) in presented successively eleven faces. T2 was three types: neutral, high, or low negative emotional valence angry faces. The results indicated that the T2 accuracies of angry faces were higher than neutral face. However, the performance did not differ between angry faces. Therefore, it is suggested that negative stimuli itself include the processing superiority regardless of the intensity of negative emotional valence. This research was supported by JSPS KAKENHI Grant Number 15H06569.

P2442 MRI-compatible wearable actuator for measuring sense of agency

Shota Tanaka, Koichi Suzumori, Hiroyuki Nabae, Takako Yoshida *Tokyo Institute of Technology, Japan*

A power assist suit is a wearable robot that augments human power and supports human action. Although demand is growing for power assist suits in the market, scientific understanding of human factors remains still unclear, such as the brain processes associated with suit usability. Here, we developed a MRI-compatible wearable actuator that generates wrist flexion and extension by pressurization of McKibben artificial muscles using 2.0 mm diameter tubes. The actuator can be controlled by solenoid valves located outside the MRI room. Our aim is to use the actuator in studies of brain processes and plasticity related to sense of agency, which is the sense that the self, not the robot, causes body movement. We examine this sense in relation to self-body recognition in order to measure and optimize the usability of power assist suits. Possible psychophysical measurements and functional MRI studies are discussed.

P2443
The time-course of sensitivity in contour detection by using fragmented object contour

Ko~suke Taniguchi, Kenji Takahashi, Miyabi Arai *Doshisha University, Japan*

Detecting the contours of objects quickly and accurately is essential for object recognition. The present study asked participants to detect fragmented object contours to investigate how shape information and exposure duration, as well as timing of response (grouped into three sets: early, middle, and late) influence the sensitivity (d') of contour detection. The contour stimuli consisted of two types of shape fragmentation (curved and straight) and two types

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of length fragmentation (short and long) similar to those used by Panis et al. (2008). The observers were asked to decide whether the stimulus presented briefly (50, 100, or 200ms) was a fragmented object contour or scrambled segments. The results showed that longer fragments and longer durations increased sensitivity. Even with longer fragments, however, delayed response decreased sensitivity. These results indicate that late reaction time rather than presentation duration disrupts accurate contour detection.

P2444
The effect of adaptation to simple line drawings on duration estimation

Tadayuki Tayama *Hokkaido University, Japan*

It has been reported that the perceived duration of high-frequency stimuli is shorter than that of low-frequency stimuli, even though the physical duration is equal. This effect could be explained by the neural inhibition hypothesis. The present study examined the hypothesis by using adaptation procedure. Simple line drawings (circle and square) were used as standard (1s), adaptation (5s), and test (0.6∼1.2s) stimuli. In one condition (a), these stimuli were presented in this order. In another condition (b), standard and adaptation stimuli were exchanged in the order. Observers estimated the duration of each test stimulus, compared with the standard stimulus by the magnitude estimation method. The results showed that durations of test stimuli in condition (a) were estimated shorter than those in the control (no adaptation) condition, if shapes of adaptation stimuli were the same as those of test stimuli. These supported the inhibition hypothesis.

P2445
Negative correlation between strength of visually induced self-motion and subsequent motion aftereffects

Wataru Teramoto *Kumamoto University, Japan*

Viewing long-lasting large-field visual motion causes not only self-motion sensation (vection) but also motion adaptation, resulting in visual motion aftereffects (MAEs). The present study investigated a relationship between vection strength and the magnitude of MAEs to clarify visual motion processing during self-motion. A virtual cylinder with a radius of 1 m was generated and stereoscopically displayed on the screens of the Oculus Rift. It was rotated around the earth-vertical axis at 30°/s for 1 min to induce circular vection and, then, kept stationary for another 1 min to measure the MAEs. Results showed that the stronger vection occurred, the shorter the MAEs lasted. This finding suggests that the self-motion mechanisms activated by large-field visual motion conversely recalibrate or bias visual motion processing, probably to stabilize the perceptual world during self-motion.

P2446
The effect of the height of the first-person perspective on bodily self-consciousness during the full-body illusion

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The bodily self-consciousness (BSC) means the sense of self and its close link to neural body representation. Among three components of BSC -self-location, self-identification, and the first-person perspective (1st PP), we focused on 1st PP. To investigate the effect of the height of the 1st PP on BSC, we modified the full-body illusion by setting the camera on the different heights and using Oculus Rift DK2. We confirmed the illusion occurs on any heights of the 1st PP, and our result showed that the illusion occurs more on the condition of the higher/lower heights than the normal height condition. Furthermore, it was shown that the higher is the 1st PP, the higher we perceive ourselves. These results suggest that our BSC highly depends on the 1st PP, that is, BSC changes according to the condition of the 1st PP.

P2447
Perceptual learning technology improves the mapping of ambiguous facial expressions to emotional category words

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Recent research suggests that perceptual learning technology, which is known as perceptual leaning modules (PLMs), could improve the identification of structural information from patterns (e. g., Kellman, Massey & Son, 2010). We investigated whether PLMs facilitated the identification of information from facial expressions in an experiment that consisted of a pretest, a PLM intervention, an immediate posttest, and a delayed posttest one week later. Two types of PLM interventions were used. Results indicated that improvements in reaction times between pretest, and delayed posttest for mapping emotional category words to facial expressions were significantly greater in the multi-direction-viewing PLM condition, compared to the single-direction-viewing PLM condition. These results suggest that viewing faces in multiple directions could improve picking up information from facial expressions and that fluency in picking up information could be transferred to novel category-discrimination tasks.

P2448
Length perception by head and by hand

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Flexibility is a fundamental hallmark of perception and behavior in all animal species. This hallmark is manifest in perception by touch. Many different object properties can be perceived by wielding an object, and a given property of a wielded object can be perceived by different configurations of the touch system. For example, perceived length is comparable when an object is wielded by one hand, both hands, the foot, or the torso. In the current experiment, participants perceived the length of six unseen rods wielded by hand or attached to a helmet. In both conditions, actual length accounted for over 95% of the variance in perceived length, and perceived length differed across conditions only for the shortest rod length. These results (1) suggest that these two appendages do not differ qualitatively with respect to length perception and (2) are consistent with characterization of the touch system as a smart perceptual instrument.

P2449
Neuronal entrainment to hierarchical visual rhythms

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When we enjoy reading a verse or dancing with our favorite piece of music, neuronal rhythms in our brain subserve perception by resonating with rhythmic structures at multiple spatiotemporal scales. Here we demonstrated analogous hierarchical entrainment effect in visual modality and revealed its link with visual perception. We superimposed delta-band (1Hz) temporal structure upon theta-band (4Hz) rapid serial visual presentation (RSVP) streams, organizing characters into successive 4-character Chinese idioms or shapes into periodically changed quadruplets. Such hierarchical organization entailed EEG power enhancement at 1Hz for both stimulus types, with the power strongly correlated with the perception of the quadruplets. Meanwhile, a broader 4Hz entrainment effect was observed regardless of whether or not the RSVP streams consisted of hierarchical structures. These findings suggest that multiplexed coding through entrainment synchronizes neuronal activity to temporally hierarchical visual rhythms, and provides an effective strategy to decouple visual rhythm perception from visual information parsing.

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P2450
Temporal distortion in the perception of actions and events

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When a sensory event triggers an action, it is perceived to have occurred later in time than an event that does not (Yabe & Goodale, 2015). This finding mirrors the intentional binding phenomenon in which a sensory event that is triggered by a voluntary action is perceived to have occurred earlier in time than an identical event that is not triggered by an action (Haggard et al., 2002). It is not clear how the perceived timing of a sensory event embedded within a longer sequence of actions and events is affected by its position in that sequence. Here we show that the sensory events are bound to the actions both in event-action and in action-event dyads which are part of a longer sequence of actions and events - and these temporal effects compete with one another in determining when the events are perceived to have occurred.

P2451
Motion perception plays a role in the Anti-Gravity Lens illusion.

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When a certain SOA is introduced between the inducer and the target of "Gravity Lens illusion" (Naito & Cole, 1994), the direction of the illusory shift of the target can be reversed ("Anti-Gravity Lens illusion"). In this study, we examine how the directional relationship between the inducer and target affects the Anti-Gravity Lens illusion. The target was displayed at an eccentricity of 5 deg and the inducer was either on the nearer or farther side of it. It was found that the amount of the illusory shift of the target was greater in the centrifugal direction (the inducer was nearer than the target to the fixation point) than in the centripetal direction (the inducer was farther than the target). This directional anisotropy is consistent with the characteristic of motion perception which prefers centrifugal motion, suggesting the contribution of the motion processing system to the Anti-Gravity Lens illusion.

P2452
Exploring the other-race effect in discrimination and categorization task with Taiwanese adults: Using a morph face paradigm

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Using morphed face images, the present study tested the own-race recognition advantage and the other-race classification advantage with two tasks in Taiwanese adults. In Experiment 1: the same/different face discrimination task, participants viewed an Asian- and Caucasian-parent face followed by either the same (0%) or a different morphed face (10%∼60%). In Experiment 2: the race categorization task, the same participants were asked to categorize a series of morphed faces (0%∼100%) as Asian or Caucasian. The results showed the presence of own-race discrimination advantage that participants had a lower threshold for the Asian-parent condition (Exp.1), and an other-race categorization advantage that the 50% morphing face (half Asian, half Caucasian) was judged as more like a "Caucasian" face (Exp.2). Furthermore, participants who categorized the 50% morphing face as "Caucasian" exhibited higher discrimination thresholds and a greater magnitude of the other-race effect, whereas those who categorized as "Asian" exhibited the opposite results.

P2453
The effect of sound on visual grouping in a multi-stable stimulus

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In this audiovisual perception study we used a stimulus (the "Polka dance" stimulus) consisting of identical bars that move in circular motion, with acceleration towards and deceleration away from the bars' point of coincidence. The stimulus can be seen in four different ways: the bars are either seen as moving through each other (streaming), as bouncing off of each other at their point of coincidence (bouncing), or as moving in unison in clockwise or counter-clockwise circular motion. Here we show that the presentation of sound affects the appearance rate of some of the four percepts. Following earlier research on the streaming-bouncing illusion, the presentation of a click at the bars' point of coincidence facilitated the bouncing percept. Auditory motion information from a click train panned between the speakers facilitated visual (counter) clockwise rotation. The results provide examples of how visual grouping and motion is affected by auditory information.

P2454
Neural entrainment to rhythmic structures reallocates visual attention over time

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Neural entrainment refers to the synchronization of brain oscillations to rhythmic stimuli, thereby providing a neural substrate for selective attention to external rhythms. In this study, we investigated how neural entrainment mediates the temporal deployment of visual attention. We presented a rapid series of visual items containing two targets, along with a synchronous stream of auditory items grouped into periodic units. The impairment in detecting the second target, known as 'attentional blink', was significantly alleviated when the two targets appeared in adjacent relative to in the same temporal units, suggesting the redeployment of attention according to the rhythmic context. More importantly, electroencephalographic recordings showed cortical entrainment tracking the structure of contextual rhythms, with significant correlation between the strength of entrainment and the magnitude of the observed attentional effect. These results altogether highlight the role of neural entrainment in reallocating temporal attention through extracting temporal regularities in the dynamic environment.

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